Chapter 4
Temperature Measurement

4.1 Principles of Temperature Measurement

Temperature measurement relies on the transfer of heat energy from the process material to the measuring device. The measuring device therefore needs to be temperature dependent.

There are two main industrial types of temperature sensors:

- Contact
- Non contact

Contact

Contact is the more common and widely used form of temperature measurement. The three main types are:

- Thermocouples
- Resistance Temperature Detectors (RTD's)
- Thermistors

These types of temperature devices all vary in electrical resistance for temperature change. The rate and proportion of change is different between the three types, and also different within the type classes.

Another less common device relies on the expansion of fluid up a capillary tube. This is where the bulk of the fluid is exposed to the process materials temperature.

Non-Contact

Temperature measurement by non-contact means is more specialised and can be performed with the following technologies:

- Infrared
- Acoustic

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4.2 Thermocouples

4.2.1 Basis of Operation

A Thermocouple consists of two wires of dissimilar metals, such as iron and constantan, electrically connected at one end. Applying heat to the junction of the two metals produces a voltage between the two wires. This voltage is called an emf (electro-motive force) and is proportional to temperature.

A thermocouple requires a reference junction, this is placed in series with the sensing junction. As the two junctions are at different temperatures a thermal emf is generated. The reference junction is used to correct the sensing junction measurement.

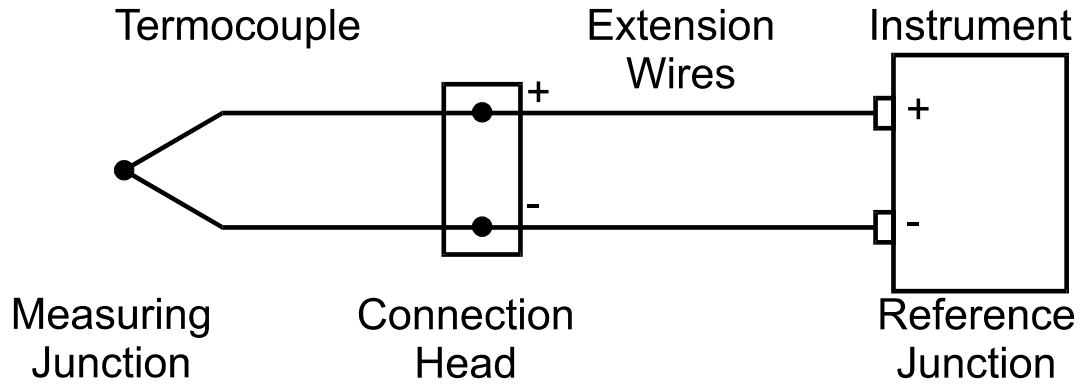


Figure 4.1
Typical thermocouple and extension leads

The voltage across the thermocouple increases as the temperature rises and a suitably calibrated instrument, capable of measuring small voltages, can be used to measure the change. The process temperature is obtained from the voltage, either by reading from a graph or by using thermocouple tables. Thermocouple tables list the voltages corresponding to each temperature. A table is required for each thermocouple type.

The relationship between millivolts and temperature is not linear. In microprocessor based equipment, the conversion is done based on the data stored in the device.

The Reference Junction:

The sensing, or hot junction is inserted into the area where the temperature is to be measured. The reference, or cold junction is normally connected to the measuring instrument and held at 0 ^{o}C.

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For accurate temperature measurement, the reference junction temperature must remain constant or suitable compensation provided if it should change. To reduce inaccuracies, most thermocouples are now installed with instruments that provide automatic reference compensation.

One of the most accurate ways of compensating for temperature change is to maintain the reference junction at 0 ^{o}C. This however is not that practical, and some form of compensation needs to be used. The technique of cold junction compensation measures the actual temperature and applies a correction to the thermocouple reading. The correction is made by adjusting the voltage by an amount equal to the difference between the actual temperature and 0 ^{o}C.

Another method of providing this compensation is to pass current through a temperature responsive resistor, which measures the variation in reference temperature and automatically provides the necessary correction by means of a voltage drop across the resistor.

Hardware compensation can be performed that uses electronic circuits that compensate for the ice-point reference. The main advantage over software compensation, is speed. The hardware response is not dependent on computation time. Hardware compensation is, however, only suited to a particular type of thermocouple. Hardware compensation is performed by using resistors whose combined temperature resistance coefficient curves match those of the voltage-temperature curves produced by the reference junctions.

The Seebeck Effect:

A thermocouple works on the Seebeck Effect. This is where (as previously mentioned) two wires of dissimilar metals are electrically connected at one end. When the junction is heated or cooled, a voltage is produced which is proportional to the temperature.

Note that the more technical term for voltage is emf, or Electro-motive force. This basically defines the electrical driving force.

The Peltier Effect:

The reverse of the Seebeck Effect is possible and can be useful. By applying a voltage and causing a current to flow between two wires of dissimilar metals, it is possible to generate a temperature difference. Because of the different electrothermal transport properties of the metals, it is found that one of the junctions will be heated and the other cooled. This process is referred to as the Peltier Effect.

Practical applications include cooling small electronic parts, or even to provide a 0°C reference junction for a thermocouple. Other applications using this principle are becoming increasingly popular for heating and refrigeration.

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4.2.2 Selection and Sizing

Isothermal Block:

Problems can arise from a mismatch of metals when connecting a thermocouple circuit to measuring equipment. Copper, being a commonly used conductor, is often inserted into the circuit. An Isothermal block is used to change from the metal used in the thermocouple circuit to those, such as copper, used in the measuring equipment.

The law of intermediate metals states that the introduction of a third metal into a thermocouple circuit will have no effect on the voltage, so long as the junctions of the third metal with the other two are at the same temperature.

The isothermal block ensures that the connections to the measuring wires are at the same temperature, and as such, errors are avoided. The isothermal block can also include the reference junction. Appropriate temperature measurements can be made for compensation.

These are seldom used anymore.

Extension cable and compensating cable:

An extension cable for thermocouples consists of the same metals as the thermocouple junction; this has the advantage of no extra emf's generated, however the cable does prove to be quite costly. Compensating cable is a suitable and cheaper alternative, which uses copper in both conductors.

Averaging temperatures:

Thermocouples may be used in parallel when sensing an area of flow. The voltage developed at the instrument is an average of the thermocouples used. To maintain accuracy, the thermocouples need to be of the same type and extension wires should be identical.

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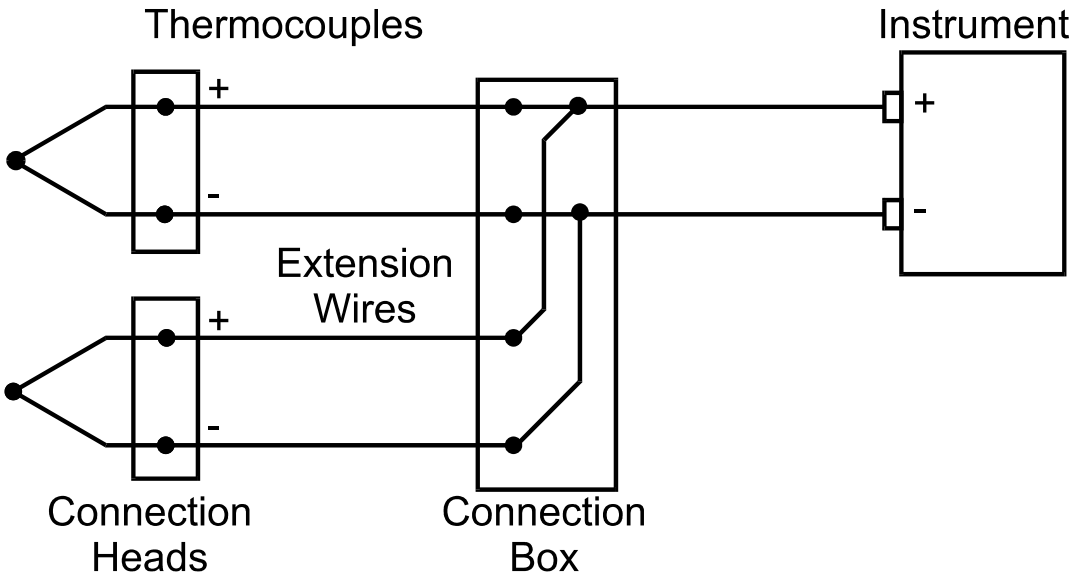


Figure 4.2
Thermocouples in parallel for average temperature measurement

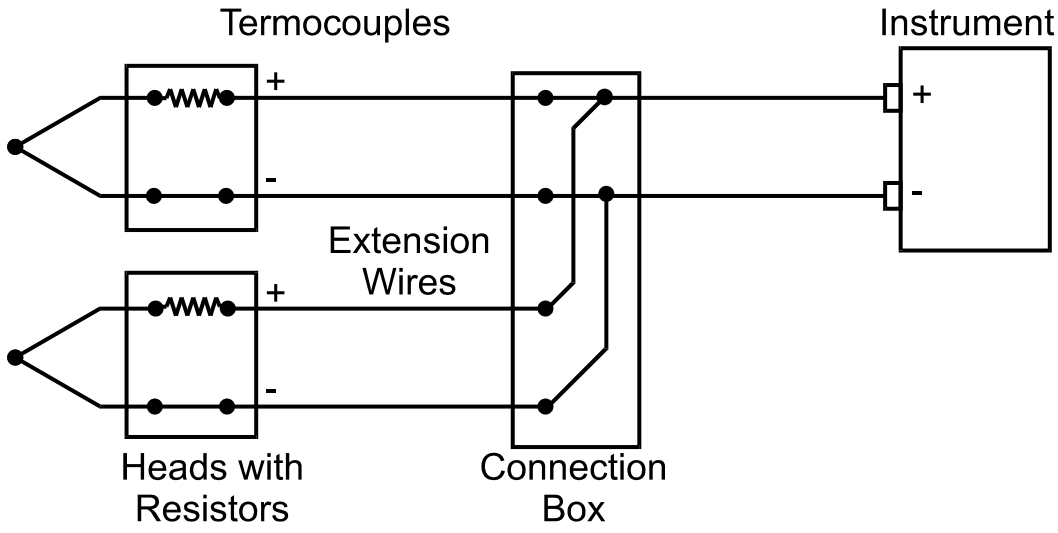


Figure 4.3
Parallel thermocouples with swamping resistors

The advantage of using thermocouples in parallel for temperature averaging is that they can be used in the same way as a single thermocouple.

When used in series for averaging temperature, the circuit requires special reference junction compensation for the increased compensating voltage, and is dependent on the number of thermocouples in the circuit. The instrument also requires calibration for the total millivolt output for the number of thermocouples used in series.

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Most thermocouples are grounded in the field. This presents a problem when using multiple thermocouples as the problems of a ground loop and shorting of signals exist.

Differential temperatures:

Two thermocouples can be used to measure the difference in temperature between two points. The thermocouples are connected together such that the voltages developed oppose each other. When measuring the same temperature, the net voltage will be zero. When a temperature difference does exist between the two points, this will be detected as a voltage and therefore the temperature measured.

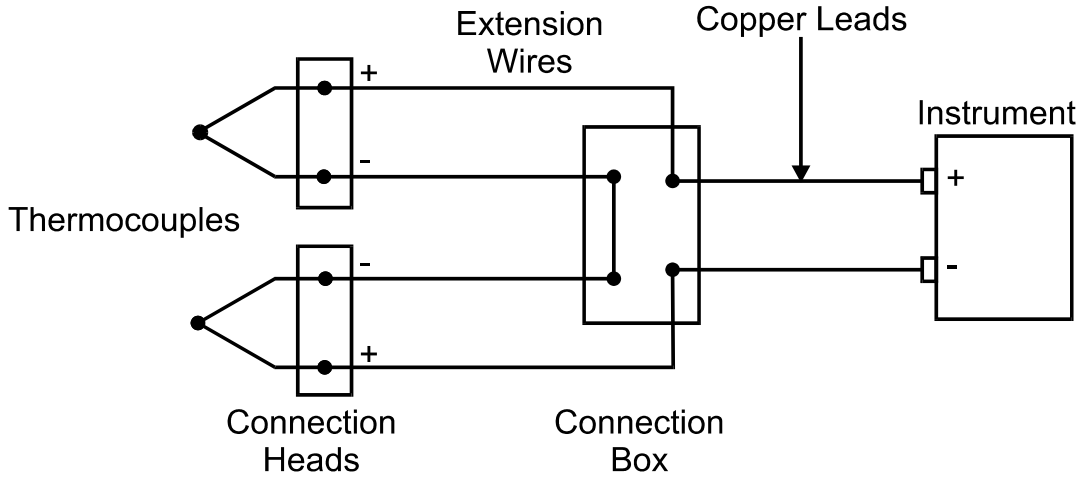


Figure 4.4
Two thermocouples used to measure temperature difference

4.2.3 Construction

Thermocouples are fusion-welded to form a pure joint, which maintains the integrity of the circuit and also provides high accuracy. Grounded junctions provide good thermal contact with protection from the environment. Ungrounded and isolated junctions provide electrical isolation from the sensor sheath.

Thermocouples are usually encased in a protective metal sheath. The sheath material can be stainless steel which is good for temperatures up to 870 ^{o}C. For temperatures up to 1150 ^{o}C Inconel is used.

A metallic oxide can be compacted into the sheath. This provides mechanical support and also electrically insulates the thermocouple junction.

The metal sheathed mineral insulated thermocouple has become the accepted norm in most industries. They use a variety of temperature and corrosion resistant sheaths and have an extremely high purity (99.4%) of Magnesium oxide insulation.

Insert Table here (Thermocouple comparison)

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Thermocouple Comparison								
ISA Type designation	Positive Wire	Negative Wire	µV/°F	Recommended Temp limits °F	Scale Linearity	Atmosphere recommended	Favourable Points	Unfavourable Points
B	Pt70-Rh30	Pt94-Rh6	3-6	32 to 3380	Same as R type	Inert or slow oxidising		
E	Chromel	Constantan	15-42	-300 to 1800	Good	Oxidising	Highest emf/°F	Larger drift than other base metal couples
J	Iron	Constantan	14-35	32 to 1500	Good; nearly linear from 300 to 800	Reducing	Most economical	Becomes brittle below 32oF
K	Chromel	Alumel	9-24	-300 to 2300	Good; most linear of all T/Cs	Oxidising	Most linear	More expensive than T or J
R	Pt87-Rh13	Platinum	3-8	32 to 3000	Good at high temps	Oxidising	Small size	More expensive
S	Pt90-Th10	Platinum	3-7	32 to 3200	Same as R	Oxidising	Same as R	More expensive
T	Copper	Constantan	8-35	-300 to 750	Good but crowded at low end	Oxidising but reducing	Resists corrosion from moisture	Limited temperature range
Y	Iron	Constantan	22-33	-200 to 1800	About same as J	Reducing		Non-standard
-	Tungsten	W74-Re26	1-12	0 to 4200	Same as R	Inert or vacuum	High temperature	Brittle; Expensive
-	W94-Re6	W74-Re26	1-10	0 to 4200	Reasonable above 60K	Inert or vacuum	Same as above	Slightly less brittle than above
-	Copper	Gold-cobalt	0.5-25	-450 to 0	Same as R		Good output at very low temp.	Expensive lab-type T/C
-	Ir40-Rh60	Iridium	1-4	0 to 3800		Inert		Brittle; Expensive

Table 4.1
Thermocouple comparison

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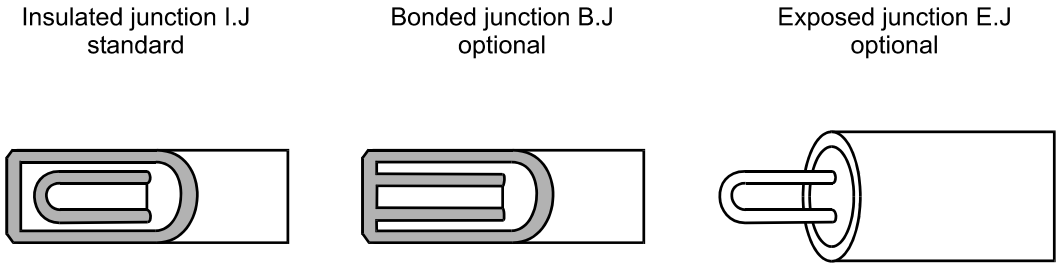


Figure 4.5
Mineral insulated thermocouples

4.2.4 Installation Techniques

Extension leads:

Care needs to be taken when extending or using long or joined leads when measuring with thermocouples. In theory, if dissimilar metals are used for connecting or extending thermocouples, and they are all at the same temperature, then no additional voltage is produced. So this should make it possible to connect a thermocouple to a measuring device as long as all components are at the same temperature.

In practice, this does not hold true. Especially with type J and T, which require extension leads of the same metal. The advantage in using consistent metals is the minimisation of effects from temperature variations as the leads pass through different thermal conditions in the plant.

The temperature variations within a thermocouple circuit are primarily dependent on the metals used and any temperature variations, and not on the length of the cable run or the diameter of the conductors.

Use the largest size thermocouple wire possible, and during installation, avoid stress and vibration. The wire length can be minimised by using transmitters that recondition the signal into a more robust and noise resistant form.

Detecting thermocouple faults:

Since thermocouples can be used in high temperature environments, it is possible that the extension wires be damaged by excessive heat. If a short develops in the wires, it may not be possible to detect. The sensing equipment will no longer be measuring the temperature at the sensing junction, but instead will measure the temperature at the short.

It should be noted, that thermocouples in this type of application are possibly used to detect high temperature fault conditions. If such a fault should occur, it is more likely to fail such that an alarm condition will not be detected. Techniques to measure the resistance continuously can be implemented to note any abrupt changes.

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4.2.5 Application Details

In a new installation, or in replacing failed equipment, it is possible to note a change in performance between the old and the new. In replacing thermocouples or RTD's there are a couple of common problems:

If the new thermocouple has been installed but does not make contact with the thermowell then an airgap is introduced which affects response time and can have a temperature variation from the actual temperature. A thermopaste can be used, and should only be applied at the tip where the temperature measurement occurs. The insertion depth is also a factor, as the deeper the insertion the more accurate the measurement. Thermopaste can make up for some shortness in length, but is limited if the shortfall is too great.

When replacing thermocouples in thermowells, it is important that the bore of the thermowell is cleaned. During the changeover or just over time, it is possible (and therefore probable!) that material may accumulate at the bottom of the well which can insulate the thermocouple from the sheath and prevent heat transfer.

Another problem is when the new thermocouple is of a different mass to the old one. This can affect the response time and, although it may not affect the accuracy, it can affect the stability in a closed loop system.

Grounding can be another problem, where the accuracy and response can differ between grounded and ungrounded devices.

Even though the thermocouple was replaced, the extension cable may require replacing also. Wear and tear or deterioration of the cable can affect accuracy. The contact with the terminals must also be considered as corrosion may be a problem.

Something that applies more to response time than accuracy is the type of junction. It may be possible that the original thermocouple had a bonded junction and the new thermocouple an insulated junction. Insulated junction thermocouples have twice the response time of bonded junction thermocouples.

4.2.6 Typical Applications

Although R-type and B-type thermocouples are suitable in oxidising atmospheres, they are easily contaminated in other atmospheres.

T-type can be used in either oxidising or reducing atmospheres. This type of thermocouple has a high resistance to corrosion due to moisture. They also provide a relatively linear output and perform well from a medium to very low temperature range.

J-type can also be used in reducing atmospheres and provide a good near-linear output. They are also the least expensive of commercially available thermocouples.

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K-type can be used in oxidising atmospheres, and are the most linear thermocouple for general use. These are the most widely used.

E-type are the most sensitive thermocouple available, and have the highest change in emf per temperature change, but they tend to drift more. They can be used in oxidising atmospheres.

4.2.7 Advantages

- Low cost
- Small size
- Robust
- Wide range of operation
- Reasonably stable
- Accurate for large temperature changes
- Provide fast response

4.2.8 Disadvantages

- Very weak output, millivolts
- Limited accuracy for small variations in temperature
- Sensitive to electrical noise
- Nonlinear
- Complicated conversion from emf to temperature

4.2.9 Application Limitations

Small temperature changes give a very small change in voltage. A platinum thermocouple, for example, will give a change of about 10microvolts for a 1 ^{o}C change in temperature.

It is because of the weak output signal from thermocouples that they are susceptible to electrical noise and are limited to applications requiring the measurement of large changes in temperature.

Thermocouples are not linear and the conversion from the generated emf to temperature is involved.

The calibration of thermocouples does change over time, and this is due to contamination, composition changes (possibly due to internal oxidation). Rapid changes in temperature may have an effect, but high temperatures definitely can affect the stability of the device. It is proven that when a K-type thermocouple is cycled to 1100°C it can vary by as much as 10%.

The integrity of the conductivity of a thermocouple has to be maintained and as such cannot be used exposed in conductive fluids.

4.2.10 Summary

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Thermocouples are the most economical temperature measuring device available and also provide measurement of the highest temperatures. The emf that they generate is independent of the wire length and diameter, however noise can become a factor.

Thermocouples provide a wide range of temperature measurement but are not recommended for narrow span or small temperature difference measurement.

For critical temperature measurement, an accurate reference junction temperature needs to measured and compensated for. This may require the use of an RTD.

If the process measurement can be performed with another device other than a thermocouple, then that should be considered. Thermocouples are low cost and suited for applications requiring a large temperature range.

4.2.11 Technical Data - Thermocouple Tables

The output of a thermocouple is quite small, in the order of millivolts. Depending on the type of thermocouple, the range can vary from -11 to 75mV. The relationship between the temperature and voltage is not linear and calibration curves or formulae are used for the conversion.

Orientation table for temperature sensors

Type	Temperature Range °F	Available Span	Accuracy % of span	Stability	Repeat-ability	Response Time	Sensit-ivity	Linear
T/Cs:								
Type T	-300 to 750	<100 to 1000	0.1	G	G	G	G	N
Type J	32 to 1500	<100 to >1000	0.1	G	G	G	G	N
Type K	-300 to 2300	<100 to >1000	0.1	G	G	G	G	-
Types R&S	32 to 3000	<1000 to >1000	0.1	G	G	E	E	N
RTDs:								
Pt	-400 to 1200	<100 to >1000	0.15	E	E	G	G,E	-
Nickel	-350 to 600	<100 to <1000	0.25	G,E	E	G	E	N
Thermistors	-100 to 500	<100 to <1000	0.2	F	G	E	E	N
Electronic	-350 to 325	<100 to <1000	0.2-2	G	G	G	E	-
Bimetallics	-100 to 1000	<100 to <1000	1-2	E	F	G	G	N
Filled:								
Liquid	-50 to 800	<100 to <1000	0.5-2	E	F	F	G	-
Vapour	0 to 600	<100 to <1000	0.5-2	E	F	G	F	N
Gas	-100 to 1800	<1000 to >1000	0.5-2	E	F	G	F	-
Mercury	-10 to 1000	<100 to >1000	0.5-2	E	F	F	G	-
Glass stem	-10 to 1000	<100 to <1000	0.1-2	E	E	F	G	-
Infrared	-10 to 6300	<1000 to >1000	1-2	F	F	E	G	N
Ultrasonic	1500 to 15000	<100 to >1000	5	G	F,G	E	E,G	N

Table 4.2
Orientation Table for Temperature Sensors

Thermocouple tables are based on a reference junction of 0 ^{o}C. If the reference junction is not at 0 ^{o}C then a correction factor must be applied.

Thermocouple tables are provided in Appendix A.

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Calculating Temperature from Voltage (reference junction = 0 ^{o}C):

1. Select the correct table for the thermocouple type in use. ie. J,S,T.

2. Locate the millivolt reading in the body of the table, and read from the margins the temperature value.

Note, that this only gives an accuracy to that of the increments on the scale, possibly 5 ^{o}C in this case. For more accurate measurement, a straight line approximation (or interpolation) can be made between two values. Interpolation takes into account the proportionate part of the difference between the two values read from the table.

$$T_M = T_L + \frac{(T_H - T_L)}{(V_H - V_L)} . (V_M - V_L)$$

V_M is the measured voltage
V_H is the higher voltage read from the table
V_L is the lower voltage read from the table
T_M is the calculated temperature
T_H is the higher temperature read from the table (corresponding to V_H)
T_L is the lower temperature read from the table (corresponding to V_L)

3. Read the values of V_H, V_L, T_H and T_L from the table. From the above, interpolate the more precise temperature.

Example:
A voltage of 14.82 mV is measured with a type J thermocouple with a 0 ^{o}C reference temperature. Find the temperature of the sensing junction.

Solution:
From the table for J-type thermocouples, it is found that V_M= 14.82mV is between V_L = 14.67mV and V_H = 14.94mV. These voltages have corresponding temperatures of T_L = 270.0 ^{o}C and T_H = 275.0 ^{o}C.

$$T_M = 270.0\ ^{o}C + \frac{(275.0\ ^{o}C - 270.0\ ^{o}C)}{(14.94mV - 14.67mV)} . (14.82mV - 14.67mV)$$

$$T_M = 272.8\ ^{o}C$$

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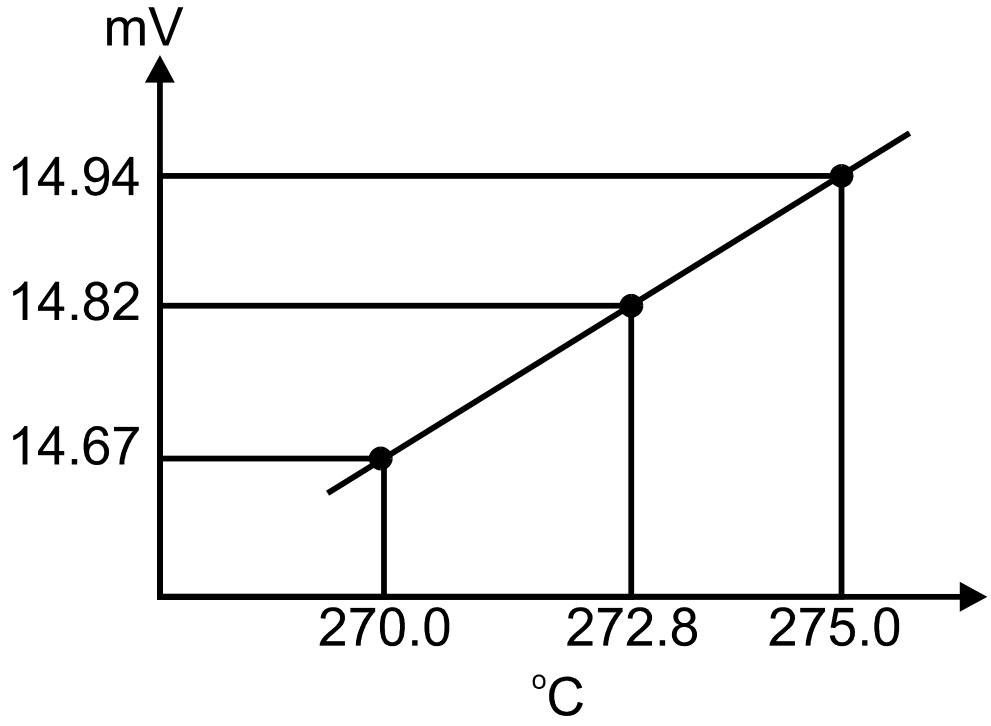


Figure 4.6
Thermocouple Interpolation

Calculating Temperature from Voltage (reference junction <> 0 °C):

1. Select the correct table for the thermocouple type in use. ie. J,S,T.

2. Locate the millivolt reading for the REFERENCE junction in the body of the table, and read from the margins the temperature value.

3. ADD this millivolt reading to that measured by the instrument.

4. This corrected millivolt reading can be converted, using the table, into measured temperature.

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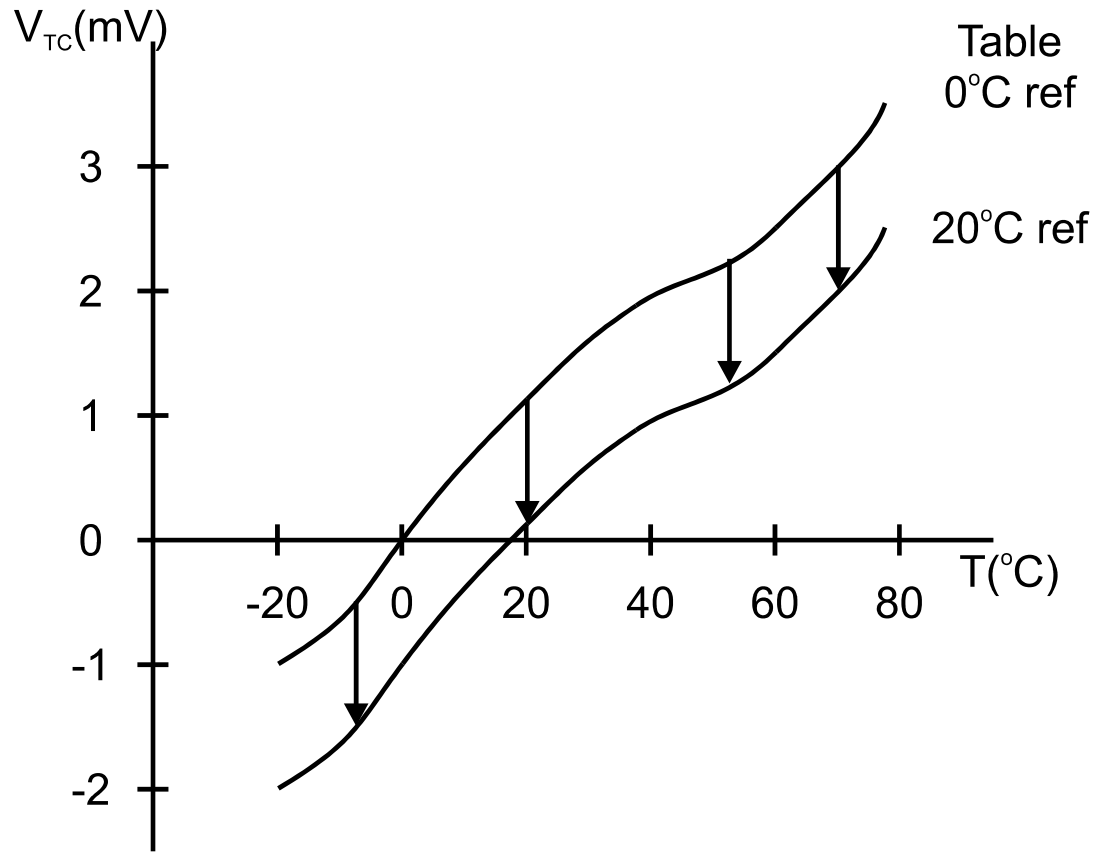


Figure 4.7
A change in reference from 0°C to 20V is equivalent to sliding the thermocouple curve down in voltage.

Example:
A voltage of 3.444 mV is measured with a type S thermocouple with a 25.0 °C reference temperature. Find the temperature of the sensing junction.

Solution:
The voltage of the reference junction from the S-type table is 0.143 mV.

The measured voltage needs to be corrected such that the actual sensing voltage is the sum of the two:
0.143 + 3.444 = 3.587mV

From the S-type table, 3.587mV gives a temperature of 435 °C.

Calculating Voltage from Temperature (reference junction = 0 °C):

1. Select the correct table for the thermocouple type in use. ie. J,S,T.

2. Locate the temperature value from the margins and read the millivolt number in the body of the table.

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Note, that this only gives an accuracy to that of the increments on the scale, possibly 5 ^{o}C in this case. For more accurate measurement, a straight line approximation (or interpolation) can be made between two values. Interpolation takes into account the proportionate part of the difference between the two values read from the table.

$$V_M = V_L + \frac{(V_H - V_L) \cdot (T_M - T_L)}{(T_H - T_L)}$$

V_M is the measured voltage
V_H is the higher voltage read from the table
V_L is the lower voltage read from the table

T_M is the calculated temperature
T_H is the higher temperature read from the table (corresponding to V_H)
T_L is the lower temperature read from the table (corresponding to V_L)

3. Read the values of V_H, V_L, T_H and T_L from the table. From the above, interpolate the more precise voltage.

Example:
A temperature of 272.8 ^{o}C is measured with a type J thermocouple with a 0 ^{o}C reference temperature. Find the output voltage for this temperature.

Solution:
From the table for J-type thermocouples, it is found that T_M= 272.8 ^{o}C is between T_L = 270.0 ^{o}C and T_H = 275.0 ^{o}C. These temperatures have corresponding voltages of V_L = 14.67mV and V_H = 14.94mV.

$$V_M = 14.67\text{mV} + \frac{(14.94\text{mV} - 14.67\text{mV})}{(275.0\ ^{o}\text{C} - 270.0\ ^{o}\text{C})} \cdot (272.8\ ^{o}\text{C} - 270.0\ ^{o}\text{C})$$

$$V_M = 14.82\text{mV}$$

Calculating Voltage from Temperature (reference junction <> 0 ^{o}C):

1. Select the correct table for the thermocouple type in use. ie. J,S,T.

2. Determine the voltage of the actual temperature from the table based on 0 ^{o}C reference junction.

3. Determine the voltage of the reference junction temperature from the table.

4. Take the difference between the two voltages in the last two steps.

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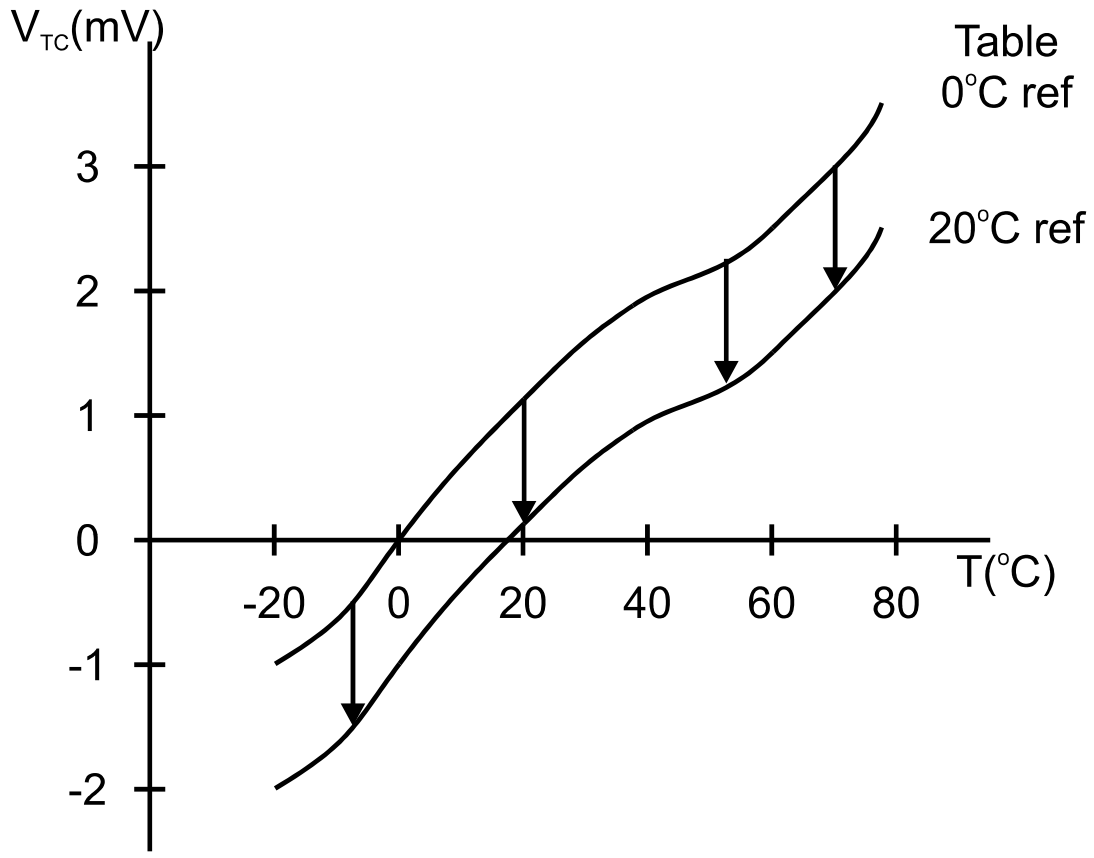


Figure 4.8
A change in reference from 0°C to 20V is equivalent to sliding the thermocouple curve down in voltage.

Example:
The calibration of an instrument is to be checked at 435 °C. This temperature is measured with a type S thermocouple with a 25.0 °C reference temperature. Calculate the millivolts expected at the instrument.

Solution:
The temperature of the reference junction is 25 °C, and from the S-type table is 0.143 mV.

The measured temperature should be 435 °C, and again from the S-type table would give a voltage of 3.587mV. Taking the reference voltage into account, the actual voltage that would need to be seen is:

3.587mV - 0.143mV = 3.444mV

For a temperature of 435 °C, with the reference at 25 °C, we would expect to see 3.444mV.

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Thermocouple Tables

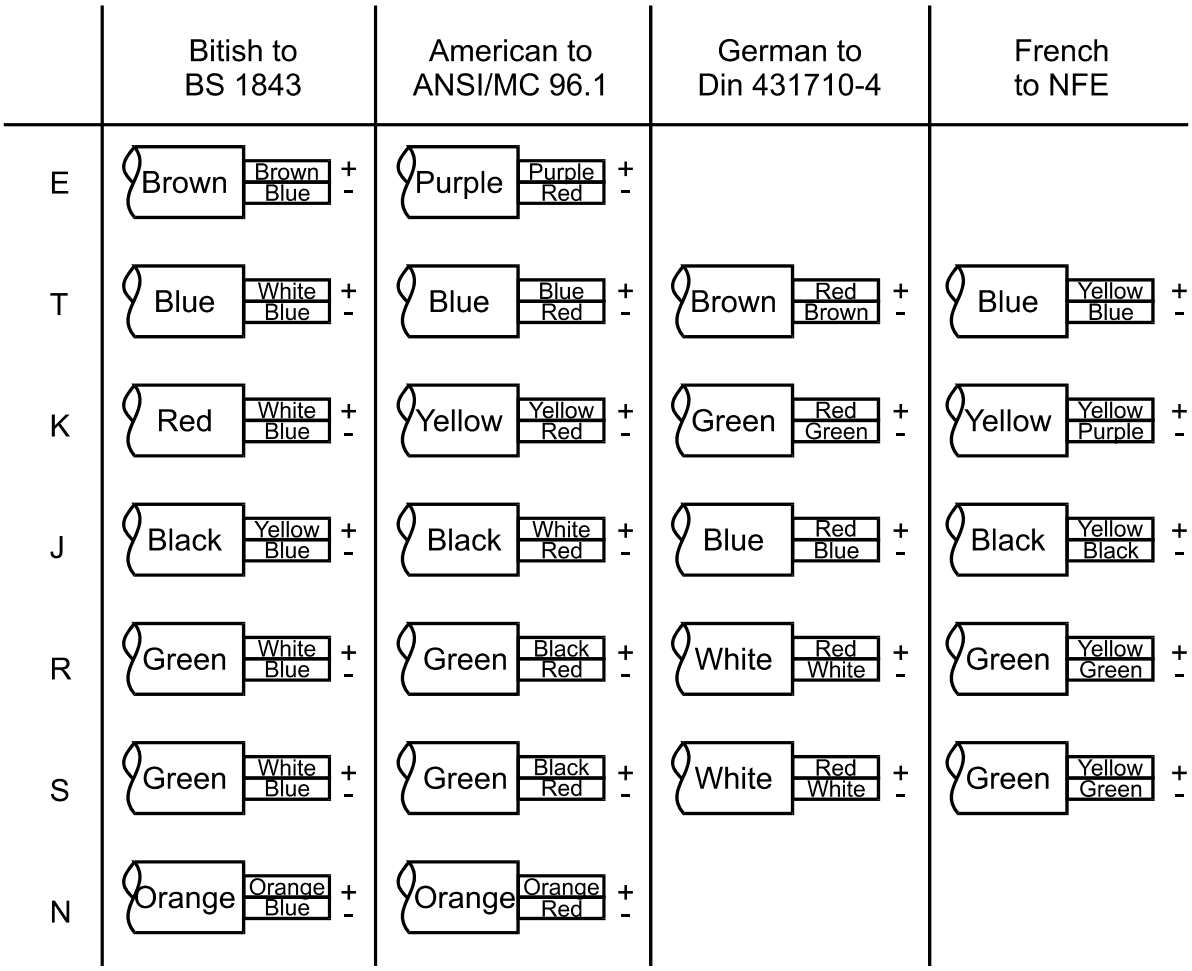


Table 4.3
Thermocouple table

Thermocouple Materials and Temperature Ranges					
Thermo-couple Type	Thermocouple Materials		Source of Data	Temperature Range	
	Positive wire	Negatve wire		Calibrated range (given in the table)	Recommended practical range
B	Platinum with 30% Rhodium	Platinum with 6% Rhodium	Hoskins Manu. Co.	+32 to +3308°F 0 to + 1820°C	+32 to +3000°F 0 to +1700°
C	Tungsten with 5% Rhenium	Tungsten with 26% Rhenium	Hoskins Manu. Co.	+32 to +4200°F 0 to +2315°C	+ 32 to +3000°F 0 to +1700°C
N	Nicrosil	Nisil	Degussa Manu. Co.	-454 to +2372°F -270 to +1300°C	+32 to +2300°F 0 to 1250°C
E	Nickel Chromium alloy ("Chromel")	Copper-Nickel allow ("Constantan")	NBS	-454 to +1832°F -270 to +1000°C	-300 to +850°F -200 to +450°C
G2	Tungsten	Tungsten with 26% Rhenium	Hoskins Manu. Co.	+32 to +4200°F 0 to +2315°C	+32 to +700°F 0 to +400°C
J	Iron	Copper-Nickel alloy ("Constantan" – SAMA version)	NBS	-346 to +2192°F -210 to 1200°C	+32 to +700°F 0 to +1000°C
K	Nickel-Chromium alloy ("Chromel")	Nickel-Aluminium alloy ("Alumel")	NBS	-454 to +2500°F -270 to +1372°C	-300 to 1600°F -200 to +900°C
PL2	"Platinel II" ** positive	"Platinel II" ** negative	Engelhard Industries	-454 to +2543°F -270 to +1395°C	+32 to +1800°F 0 to +1000°C
R	Platimum with 13% Rhodium	Platinum	NBS	-58 to 3213°F -50 to 1767°C	+32 to +2700°F 0 to +1500°C
S	Platinum with 10% Rhodium	Platinum	NBS	-58 to 3213°F -50 to 1767°C	+32 to +2700°F 0 to +1500°C
T	Copper	Copper-Nickel alloy ("Constantan")	NBS	-454 to +752°F -270 to +400°C	-450 to +400°F -270 to +200°C

Table 4.4
Thermocouple materials and temperature ranges

4.3 Resistance Temperature Detectors (RTD's)

4.3.1 Basis of Operation

RTD's are built from selected metals (typically Platinum), which change resistance with temperature change.

The transducer is the temperature sensitive resistor itself, with the sensor being a combination of the transducer and electronics that measure the resistance of the device.

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The resistance temperature detector (RTD) measures the electrical conductivity as it varies with temperature. The electrical resistance generally increases with temperature, and the device is defined as having a positive temperature coefficient. The magnitude of the temperature coefficient determines the sensitivity of the RTD.

Apart from Platinum, other metals are used for RTD's such as Copper and Nickel. Platinum is the most common and has the best linear characteristics of the three, although Nickel has a higher temperature coefficient giving it greater sensitivity.

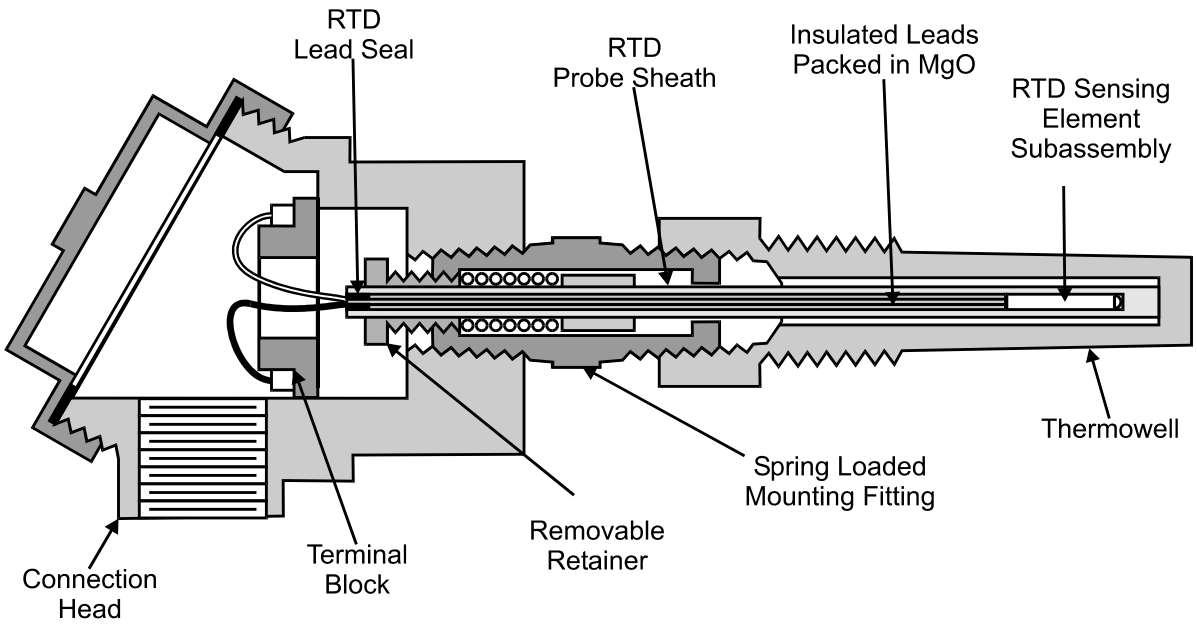


Figure 4.9
Typical RTD and thermowell construction

Temperature Coefficient:

The temperature coefficient defines how much the resistance will change for a change in temperature, and has units of ohms/^{o}C. The greater the temperature coefficient, the more the resistance will change for a given change in temperature. This ultimately defines how sensitive the device is.

RTD's are generally quite linear, however the temperature coefficient does vary over the range of operation. As an indication, the temperature coefficient for Platinum is averaged at 0.00385 over the range from 0°C to 100°C, but varies by about 2% over this range.

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4.3.2 Selection and Sizing

There are two basic types of RTD's:

- PT100
- PT1000

PT100

The 'PT' defines that the metal is Platinum and the '100' is the resistance in ohms at ice point (or 0 ^{o}C). These are generally wire wound and are quite common.

PT1000

Again, the 'PT' defines a Platinum metal as the sensing element, but a resistance of 1000 ohms can be measured at 0 ^{o}C. These are generally thin film devices and are more expensive.

200 and 500 ohm Platinum RTD's are available, but are more expensive and less common.

Platinum is most popular for RTD's, it has good calibrated accuracy, is quite stable and has good repeatability, but is quite expensive. They are, however, not as sensitive as the Nickel and Balco devices. Nickel is not quite as repeatable, but is less expensive.

The effective range of RTD's depends principally on the type of wire used as the active element. A Platinum RTD may have a range from -100 ^{o}C to 650 ^{o}C, whereas a Nickel RTD typically ranges from -180 ^{o}C to 300 ^{o}C.

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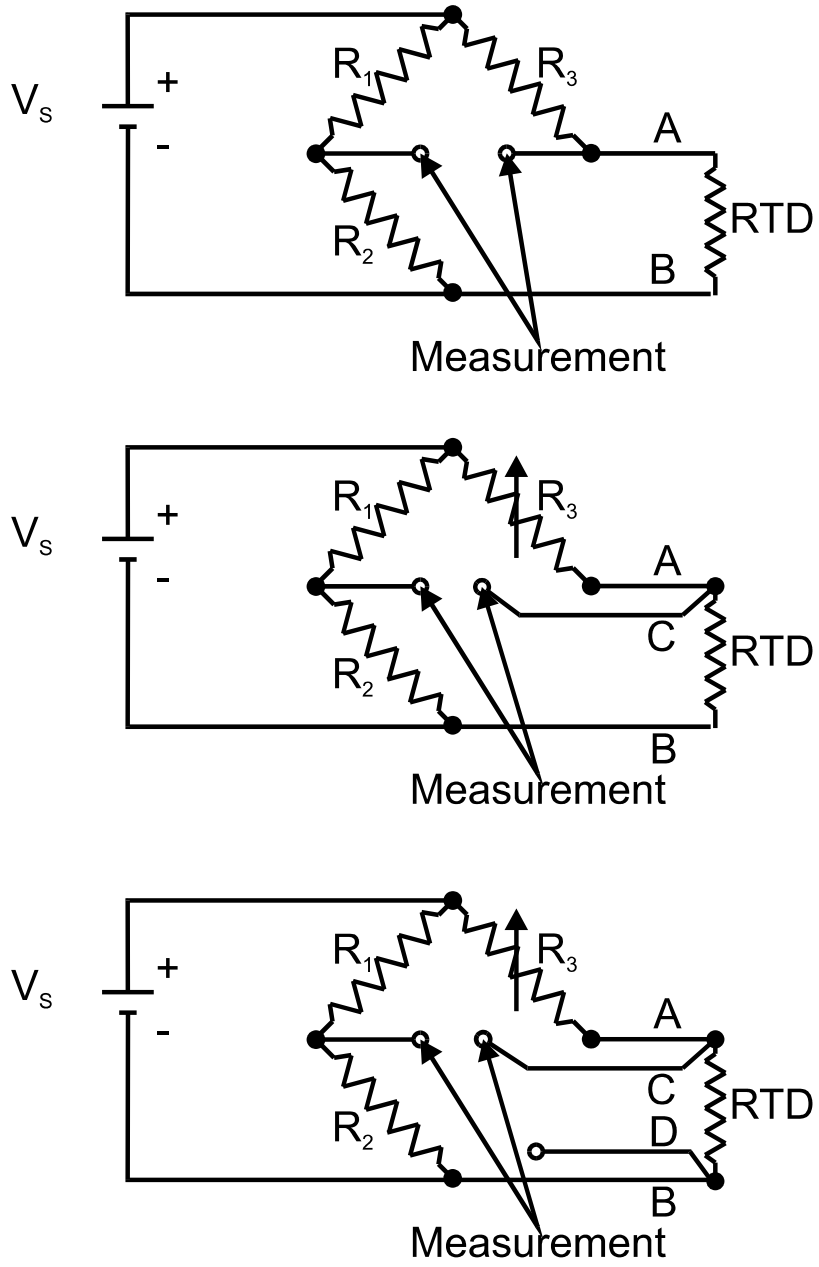


Figure 4.10
Two, three and four wire measurement

Measurement - THE WHEATSTONE BRIDGE:

There are a number of ways to measure the RTD resistance. The most common way is to use a Wheatstone bridge.

The Wheatstone Bridge consists of a bridge of three resistors located in the instrument housing, with the fourth resistor being that of the RTD. In a balanced situation, the balancing resistor is adjusted to give zero voltage across the bridge. In an unbalanced configuration, the voltage is measured across the bridge.

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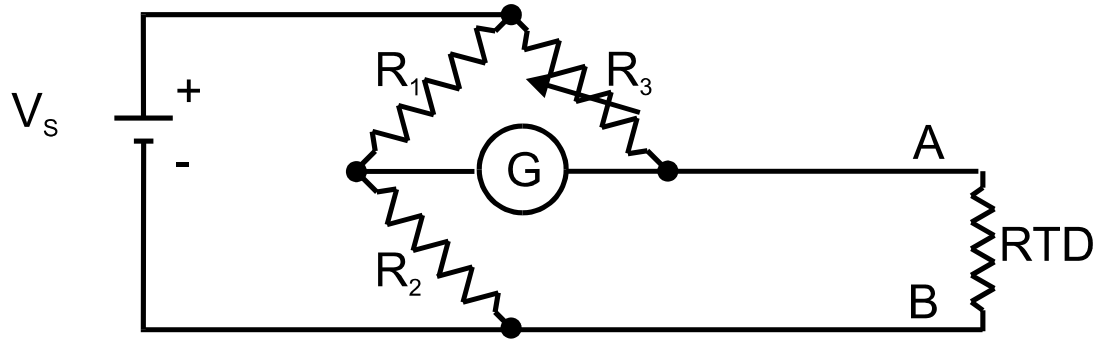


Figure 4.11
Null-balance – bridge – type two-wire RTD installation showing a galvanometric (G) readout in a balanced condition

In a bridge arrangement, the measurement of the RTD also includes the resistance of the sensing leads. There are three ways to connect up the RTD to a Wheatstone bridge, with the more complex having greater success in overcoming lead resistance problems.

RTD connection to a Wheatstone Bridge:

- Two-wire
- Three-wire
- Four-wire

Two-wire measurement:

This is the most basic type of connection for an RTD device. It is used in very simple, cheap applications. They minimise cost at the expense of accuracy. The main problem with two wire measurement is that there is no accounting for the resistance, or even change of resistance in the sensing leads. The measuring device cannot differentiate between the RTD resistance and lead resistance.

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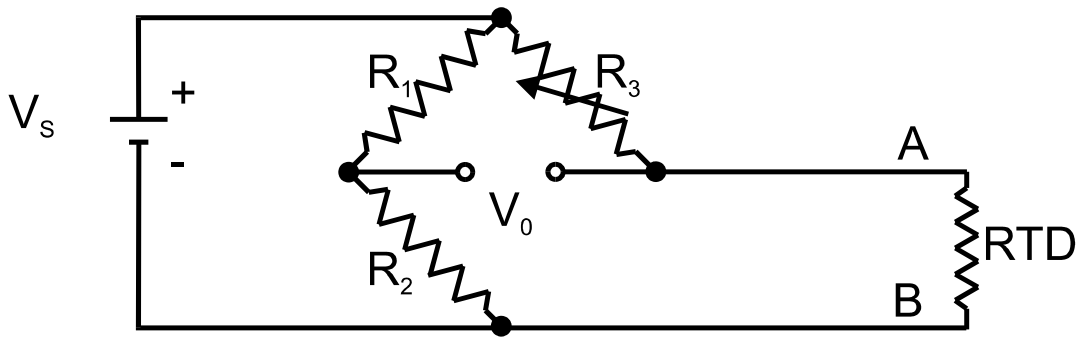


Figure 4.12
Null-balanced – bridge type two-wire RTD installation showing a digital voltmeter readout in an unbalanced condition where the DVM reads Vo volts.

When the bridge is balanced,

R3 = A + B + RTD

Without the benefits of three-wire sensing, a two-wire element can be used with a three-wire transmitter. This can reduce inventory, but is mentioned here if the need arises for system standardisation and highlights the compatibility between two and three-wire sensors. This is of particular importance for installed equipment.

Three-wire measurement:

Three-wire measurement with an RTD device balances the resistances in the lead wires within the bridge. Even though this is a simple modification to the two-wire device, it has the added cost of requiring three wires to obtain the measurement.

The concept of operation is quite simple in that one lead is measured in the top half of the bridge, with the other lead in the bottom half. Since the sensing distance and other effects are the same, the lead resistance from both sensing leads cancel.

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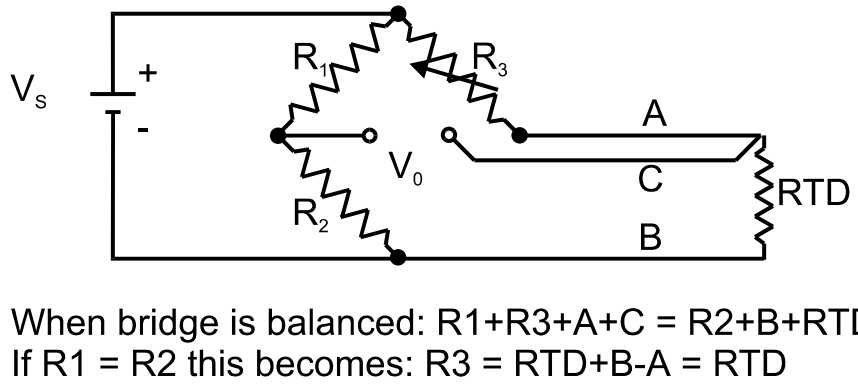


Figure 4.13
In three-wire null-balance bridge the lead-wire effect is reduced to the difference between the resistance of the two lead-wires (A - B)

When the bridge is balanced,

R3 + A + C = C + RTD + B

As A and B are identical for practical purposes,

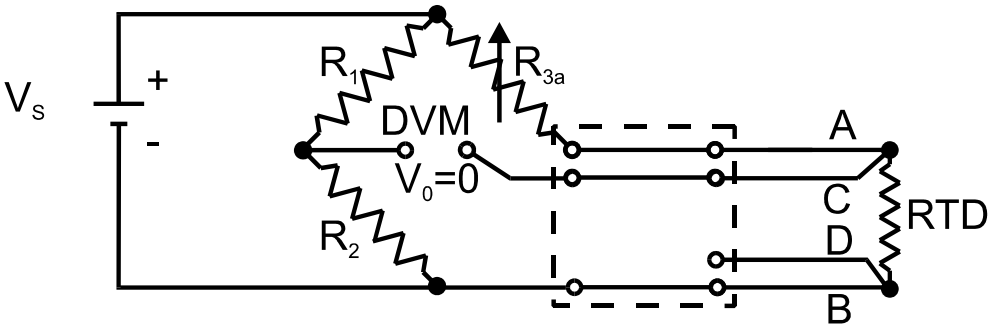
R3 = RTD

Four-wire measurement - Switched:

One of the limitations with the three-wire measurement, is that if the lead resistance is not the same or suffer different effects, then the measurement will be erroneous. The Four-wire measurement takes both sensing leads into account and alternates the leads into the upper part of the bridge.

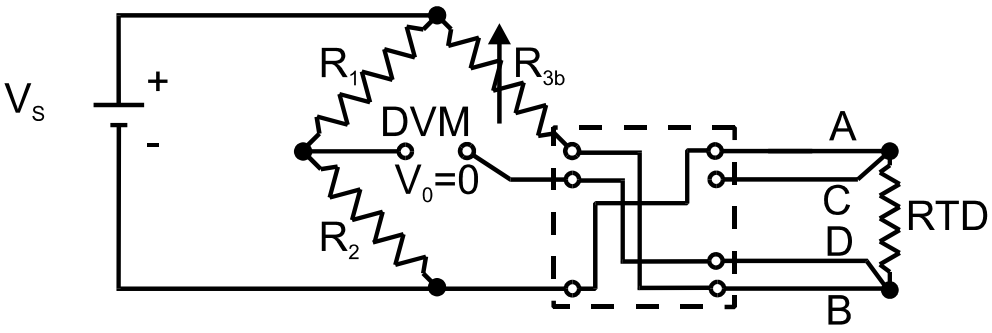
By alternating, the lead resistance is effectively measured in both sensing leads, but is then cancelled out by taking the average of the two readings. This level of complexity does make four-wire sensing more expensive.

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Switch in position "A"

R1+R3a+A+C = R2+B+RTD+C
when R1 = R2:
R3a+A = B+RTD



Switch in position "B"

R1+R3b+B+D = R2+A+RTD+D
when R1 = R2:
R3a+B = A+RTD

Figure 4.14
If the leads of a four-wire null-balance bridge are switched as shown, and the resulting two equations are added up, the lead wire effects are eliminated and the resistance of RTD = (R_{3a} + R_{3b})/2.

Four-wire measurement - Constant Current:

A simpler and more common way to eliminate errors due to lead resistance is to excite the RTD with a constant current. The main errors in sensing are due to the voltage drop caused by the excitation power in the excitation leads. The measurement of the voltage is performed by the other pair of leads which have very little current passed through them. The excitation power varies according to the manufacturers design, and is anywhere from 1 to 2mA.

The voltage drop in the power leads is defined by,

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V = IR.

Power dissipatioin is defined by,

($P = I^2R$).

As the sensing leads have very little current, there is negligible voltage drop.

Lower excitation power can be used to avoid self-heating errors.

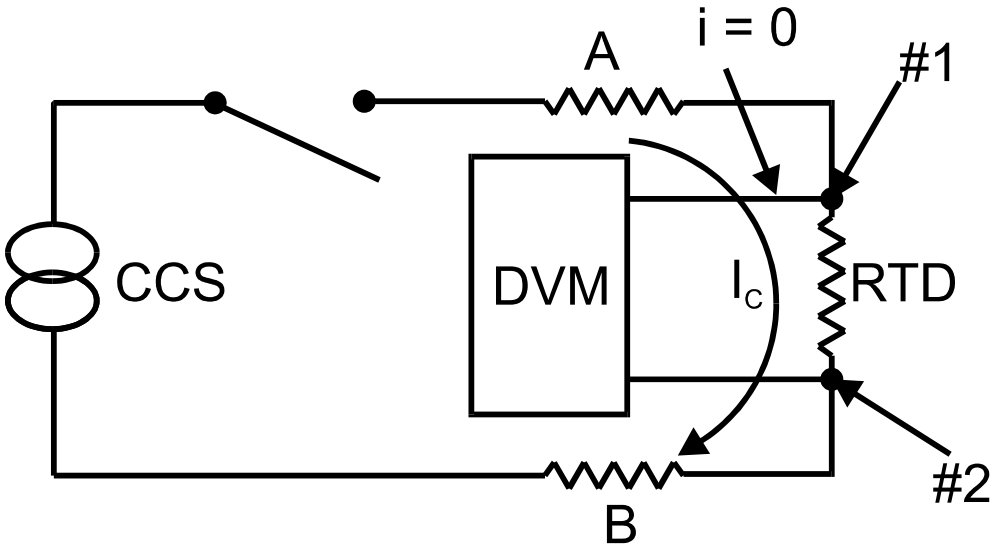


Figure 4.15
Offset compensated four-wire RTD measurement using constant voltage source and digital voltmeter as a readout (courtesy of Hewlett Packard)

4.3.3 Installation Techniques

Most of the installation techniques are common between various temperature measurement equipment. It is detailed at the end of the chapter in Installation considerations.

4.3.4 Typical Applications

A thermometer generally consists of an insert containing a PT100 housed in an external protective pipe called a thermowell. The terminal head houses either the electrical connections, or a locally mounted transmitter.

Two-wire RTD's are generally used in HVAC applications, whereas three-wire RTD's are commonly found in industrial situations. Four-wire RTD's are used in high-precision services requiring extremely good accuracy.

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4.3.5 Advantages

- Good sensitivity
- Uses standard copper wire
- Copper RTD's minimise thermocouple effect

4.3.6 Disadvantages

- Bulky in size and fragile
- Slow thermal response time due to bulk
- Self heating problems
- More susceptible to electrical noise
- More expensive to test and diagnose

4.3.7 Application Limitations

RTD's can be quite bulky, which can inhibit their use in applications. Thin film designs overcome this limitation and make this device suitable for miniaturisation.

Self heating can be a problem with RTD's. The magnitude of the errors generated by self heating effects vary, but are dependent on the size and the resistance of the RTD. These errors can be reduced by heat transfer and by minimising the excitation current.

The response time of RTD's is typically anywhere from 0.5 sec to 5 seconds. The slowness of response is due primarily to the slowness of the thermal conductivity in bringing the device to the same temperature as the surrounds. The response time increases for increased sensor size, also the use of thermowells can double the response time.

For a 2.5 mm probe the response time is 1-2 seconds, this varies with an 8mm probe having a response time of 5-10 seconds.

4.3.8 Summary

RTD's are a little more expensive but are quite stable. They are also very linear, which makes for an easier conversion between the sensing voltage and measured temperature.

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Comparison of RTD Types				
Evaluation criteria	**Platinum RTD 100Ω wire wound and thin film**	**Platinum RTD 1000Ω thin film**	**Nickel RTD 1000Ω wire wound**	**Balco RTD 2000Ω wire wound**
Cost –OEM quantity	High	Low	Medium	Medium
Temperature range	Wide -400 to 1200°F (-240 to 649°C)	Wide -320 to 1000°F (-196 to 538°C)	Medium -350 to 600°F (-212 to 316°C)	Short -100 to 400°F (-73 to 204°C)
Interchangeability	Excellent	Excellent	Fair	Fair
Long term stability	Good	Good	Fair	Fair
Accuracy	High	High	Medium	Low
Repeatability	Excellent	Excellent	Good	Fair
Sensitivity (output)	Medium	High	High	Very high
Response	Medium	Medium to fast	Medium	Medium
Linearity	Good	Good	Fair	Fair
Self-heating	Very low to low	Medium	Medium	Medium
Point (end) sensitivity	Fair	Good	Poor	Poor
Lead effect	Medium	Low	Low	Low
Physical size / packaging	Small to medium	Small to large	Large	Large

Table 4.5
Comparison of RTD Types

4.4 Thermistors

4.4.1 Basis of Operation

A thermistor is a semiconductor device formed from metal oxides. The principle of temperature measurement with a thermistor is that its resistance changes with temperature. Most thermistors differ from normal resistors in that they have a negative coefficient of resistance, this means that the resistance decreases with an increase in temperature. Negative (NTC) thermistors are the more common although positive (PTC) are also available.

Thermistors were traditionally quite unstable, and their performance was unpredictable. However, improvements in technology and the selection of oxides now means that thermistors are produced that have characteristics that are well defined.

A thermistor is a bulk semiconductor device, and as such can be fabricated in many forms. The more common include discs, beads and rods. Size does vary from a bead

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of 1mm to a disc of several centimetres in diameter and thickness. Wide ranges of thermistors (both resistance and temperature) are supplied by manufacturers. This is done by varying the doping and semiconductor materials.

4.4.2 Selection and Sizing

Types of thermistors vary in a number of ways and one change is their response to temperature changes. Thermistors are not linear, and their response curves vary for the different types. Some thermistors have a near linear temperature resistance relationship, others are available with a sharp change in slope (sensitivity) at a particular characteristic temperature.

If the range of operation is suited, it is possible to replace an RTD with a thermistor, but problems lie in obtaining conversion units that fit the desired characteristic curve and meet the required accuracy.

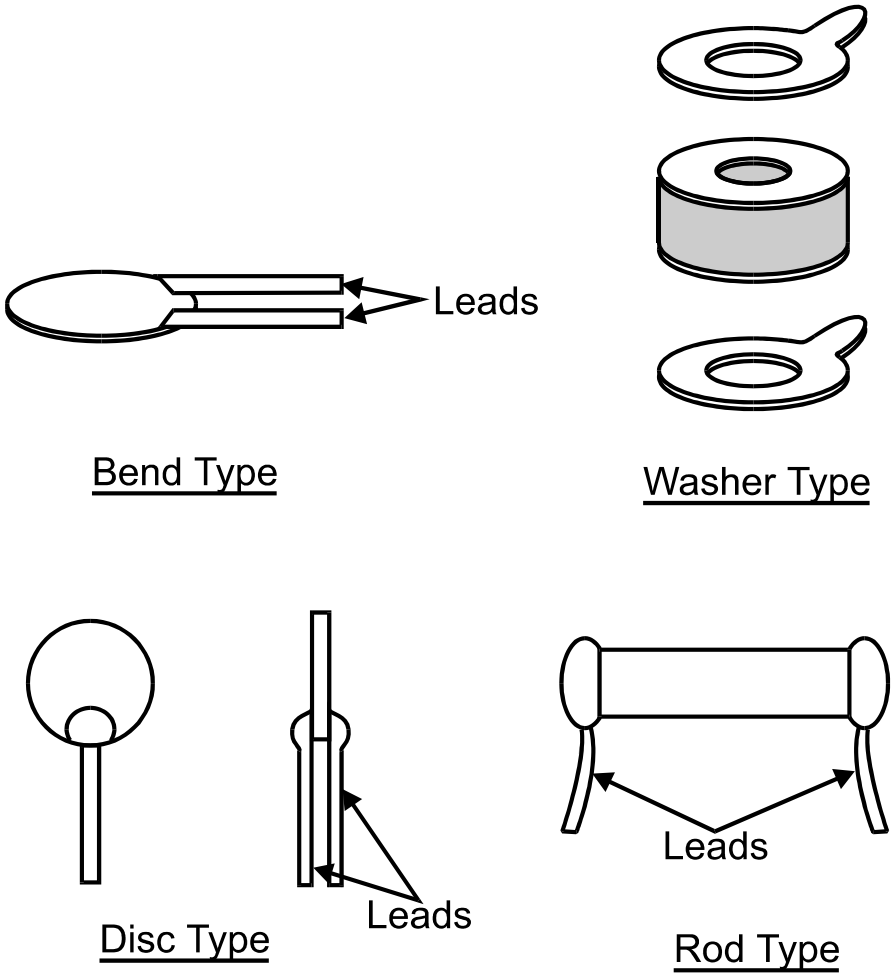


Figure 4.16
Variations in thermistor sensor packaging

Thermistors have a much higher temperature coefficient than RTD's, so a small temperature change is easier to detect. However, thermistors do not have the accuracy of RTD's and this probably accounts for thermistors being limited in

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process instrumentation. They are available in a large range of resistivities, with varying linearity.

Thermistors are available that perform temperature measurement from -73 to 316 ^{o}C (-100 to 600 ^{o}F). It should be noted that many have limited ranges and cannot be used above 120 ^{o}C (250 ^{o}F).

Some thermistors have a large change in resistance for a change in temperature. Selecting these types makes for very good narrow span measurement, however they can cause problems for wide span applications.

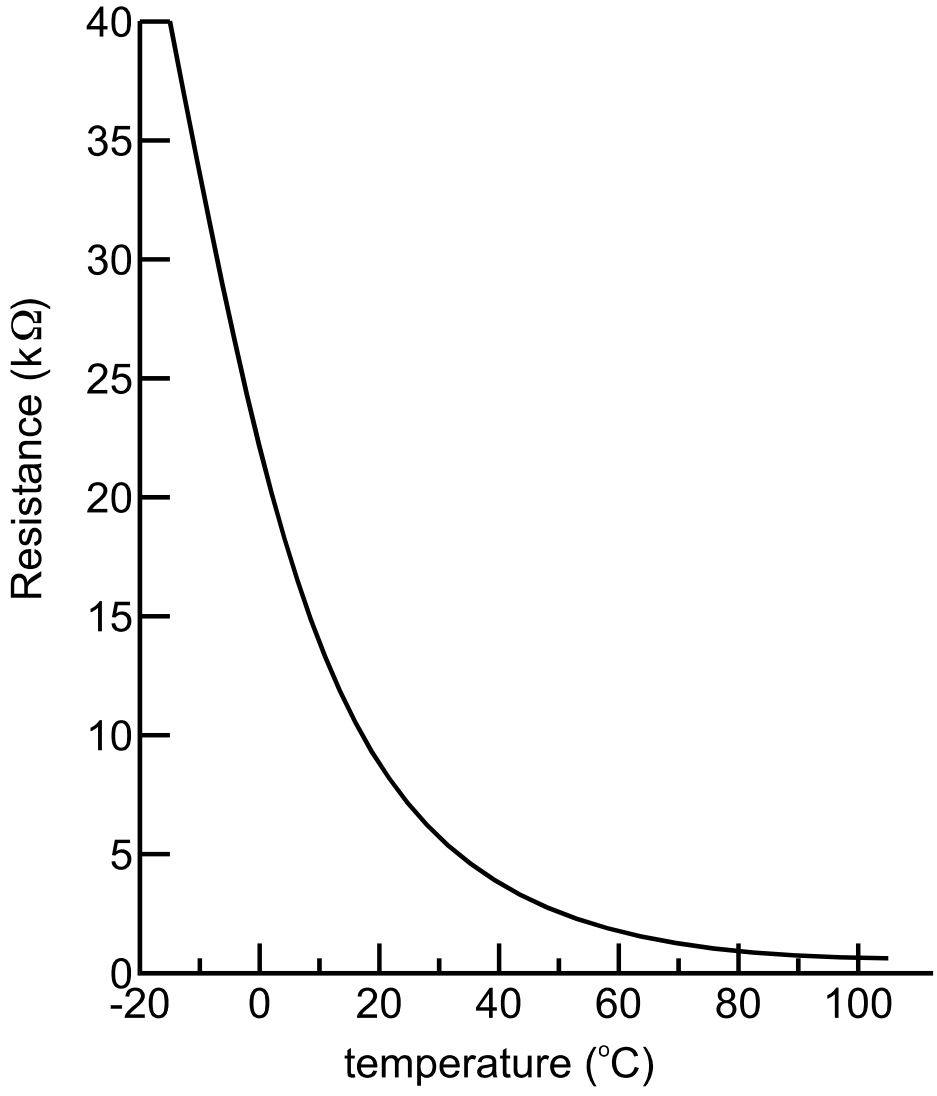


Figure 4.17
Thermistor resistance versus temperature is highly non-linear and usually has a negative slope

4.4.3 Typical Applications

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Due to their low cost, thermistors are used in many applications requiring information about process equipment for alarming and indication purposes. Such information, for example, would be the winding temperature of motors, or the temperature of bearings, or even the temperature of transformer windings.

In such applications, the actual temperature may not be of as much concern as the change or rapid increase in temperature. It is for this reason that thermistors give vital alarming information about the state of equipment in motion. In the examples of motor winding and bearings, both show a significant increase in temperature at the onset of the failure due to insulation breakdown in the windings or excessive friction in the bearings.

Application	***Cause of fault***	***Indication***
Motor winding temperature.	Breakdown of winding insulation due to moisture in windings or high motor currents (excessive starts)	Significant rise in temperature over time
Roller bearing	Excessive friction due to general wear over time, excessive loading or impulse loading	Sudden increase in temperature
Transformer windings	Breakdown of winding insulation	Significant rise in temperature over time

Table 4.6
Troubleshooting suggestions

4.4.4 Advantages

- Small size
- Fast response
- Very high sensitivity (Select range)
- No cold junction compensation
- Inexpensive
- Polarity insensitive
- Wide selection of sensors

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4.4.5 Disadvantages

- Unstable due to drift and decalibration (especially at high temperatures)
- Not easily interchangeable
- Non linear
- Narrow span
- Fragile
- High resistance, noise problems

4.4.6 Application Limitations

For most applications, self-heating is not considered a problem as the thermistor currents used are relatively low. An adjustment in the form of an offset may be made for self-heating effects if a larger current is used, but only under fixed conditions. However, anything that affects the rate of heat dissipation will change the offset required. Such affects may be changes in flow or fluid composition.

The power dissipation from a thermistor can vary, and in industrial applications, a few degrees of self-heating can be expected for each milliwatt of resistive heating. As thermistors are made smaller to cater for increases in applications, their response times become faster as is required, however the problem of the effect of self-heating also rises.

Due to the high resistance of the thermistor, consideration is required to be given to cabling, filters and even the DC voltage supply.

Range Limitations:

The range of operation of the thermistor depends on the materials used to construct and protect the sensor. There are four main limitations that affect the range of effective operation:

1) Melting or deterioration of the semiconductor.

   The semiconductor material may melt or deteriorate at higher temperatures. This condition generally limits the upper temperature to around 300 ^{o}C.

2) Deterioration of encapsulation material

   It is quite common for the encapsulating material to be plastic, epoxy, Teflon or some other inert material. This is required as it protects the thermistor from the environment and adds mechanical strength, however the material may place an upper limit on the temperature at which the sensor can be used.

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3) Insensitivity at higher temperatures

At higher temperature, the slope of the response curve is quite shallow, and possibly zero. This means that the device is physically not able to measure changes in temperature, because there is no change in resistance.

4) Difficulty in measurement at low temperatures.

As the temperature drops, the resistance of thermistors rise to quite high levels. It is because of these high resistances that low temperature limits are imposed due to the difficulty in measurement. The lower limit can be anywhere from -50 ^{o}C to -100°C.

4.4.7 Summary

It is the small size of thermistors that make this type of temperature measurement easy to implement. They are generally used for preventative maintenance applications, where the temperature goes into an alarm threshold.

Because of their non linearity and instability, thermistors are seldom used in continuous control applications.

The resistances of thermistors are a function of absolute temperature, which give the added advantage that they do not require cold junction compensation.

The stability of thermistors increases with age. Although they require recalibration due to drift, this becomes less of a problem over time as the amount of drift decreases. However, if a thermistor is operated at temperatures far below the maximum then it can remain a very stable device.

4.5 Liquid-In-Glass, Filled, Bimetallic

4.5.1 Glass-Stem Thermometers

The glass-stem thermometer is one of the oldest devices for measuring temperature. Either Mercury or alcohol is used to fill the glass bulb and extends into the capillary tube as it expands with temperature. The capillary tube where expansion occurs is evacuated, although in some applications it may be filled with nitrogen to increase the temperature range.

The material used in a glass-stem thermometer needs to remain a liquid over the full range of operation of the device. Mercury is quite common and has a large range of operation, typically -40 ^{o}C to 540 ^{o}C.

Alcohol is used for lower temperatures. As alcohol is clear, colourfast dyes are used to increase the liquids visibility.

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To increase the usability of these devices, they are mounted in metal thermowells for greater mechanical protection. All the limitations of using thermowells apply.

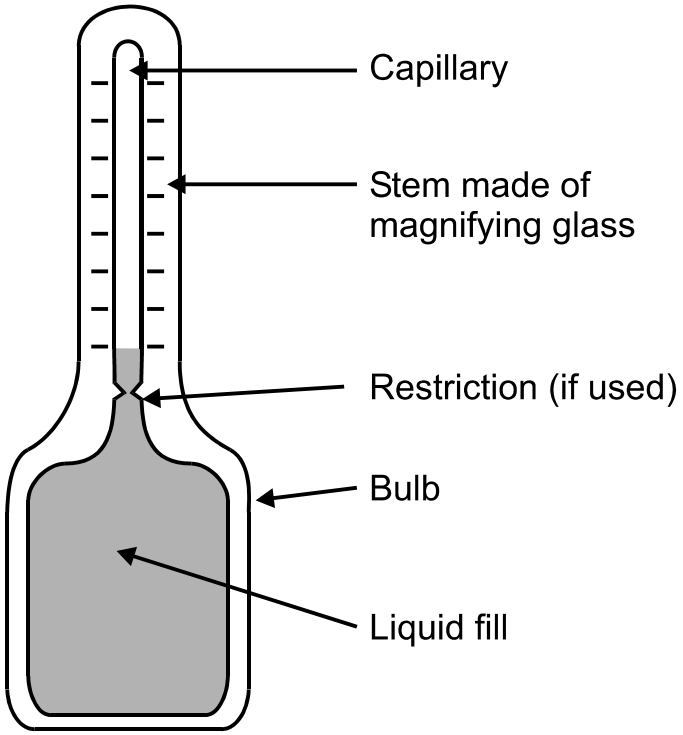


Figure 4.18
Liquid-in-glass thermometer

Advantages

- Low cost
- Simplicity
- No recalibration required

Disadvantages

- Interpretation of measurement
- Localised measurement only
- Isolated from other control and recording equipment
- Fragile

Summary

Glass-stem thermometers are seldom used anymore because of the in-accessibility of the measurement information from a control system and also the toxicity of Mercury. They are a very cheap form of temperature indication and may still be found in older systems.

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4.5.2 Filled Devices

Basis of Operation

Filled thermal systems, consist of bulbs connected through capillary tubing to pressure or volume sensitive elements. They work on the expansion of a fluid with temperature. As the temperature increases, the fluid is pushed further along the capillary, and the pressure in increased. The Bourdon type is commonly used to measure the pressure.

Selection and Sizing

In industrial applications, filled thermal systems not using glass are more common. These work on the same principle, where a bulb is filled with fluid which extends along a metal capillary tube. A pressure measuring device is used to detect the amount of expansion of the fluid.

Filled systems are classified on their expanding material, which can be a liquid, vapour, gas or mercury.

Liquid filled systems (Class I):

Liquid expansion systems have narrow spans, small sensors and give high accuracy. They also have the ability to provide temperature compensation using either an auxiliary capillary or bimetallic techniques. Fully compensated liquid expansion systems are expensive and complex.

Liquid filled systems have the advantage over Mercury in that the expansion of the fluid is about six times that of a Mercury systems. These also have the added advantage of using smaller bulbs.

The normal operating minimum for this type of sensing is from -75 ^{o}C to -210 ^{o}C, with the maximum being up to 315 ^{o}C.

Overrange protection is of particular concern in liquid filled systems, and is typically 100% over the normal operating range.

Vapour filled systems (Class II):

Vapour pressure systems are quite accurate and reliable. They also do not require any compensation for temperature effects. This form of measurement is based on the vapour-pressure curves of the fluid and measurement occurs at the transition between the liquid and vapour phases. This transition occurs in the bulb, and will move slightly with temperature, but it is the pressure that is affected and causes the measurement.

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If the temperature is raised, more liquid will vaporise and the pressure will increase. A decrease in temperature will result in condensation of some of the vapour, and the pressure will decrease.

Different materials have different vapour pressure-temperature characteristics. Methyl chloride is quite commonly used in this type of sensor. Ethane is used for low temperature operation, typically from about -70 ^{o}C to 30 ^{o}C. Whereas for high temperature applications, Ethyl Chloride can be used, with an effective operating range from 40 ^{o}C to 175 ^{o}C.

Vapour filled thermal systems are non-linear, and are generally more sensitive at the higher end of the scale. Selecting and sizing the application so that the range of operation is at the higher end can prove advantageous and provide better measurements. Overrange protection may become a problem as vapour filled systems have a low overrange limit.

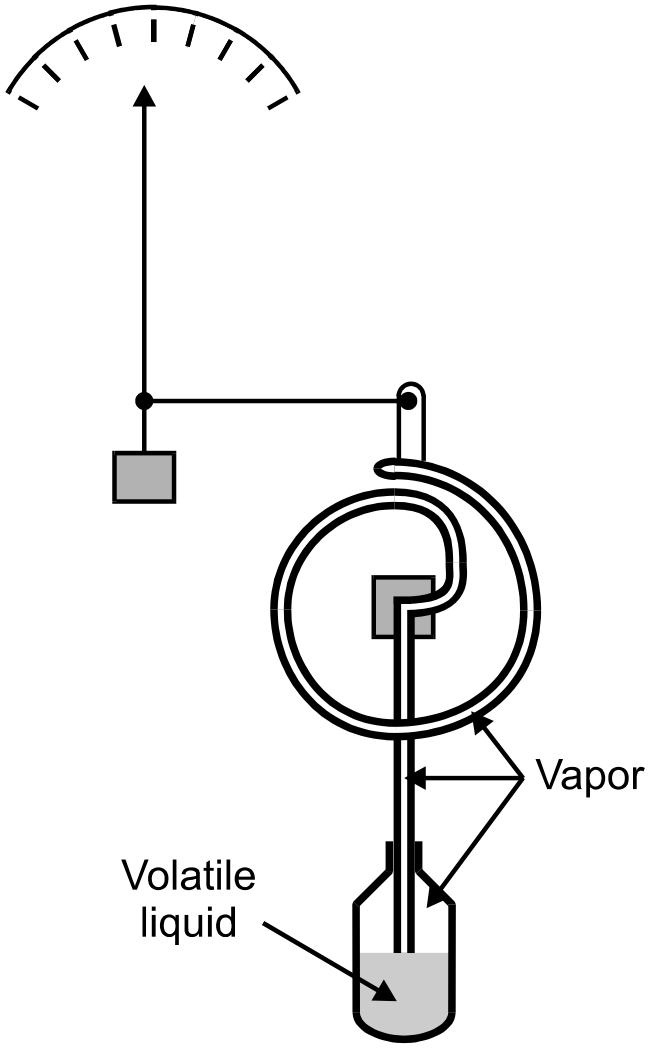


Figure 4.19
Class IIB-type Vapour Filling

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Gas filled systems (Class III):

As the volume is kept constant, the pressure varies in direct proportion to the absolute temperature. This type of measurement is quite simple and low cost.

Nitrogen is quite commonly used as it doesn't react easily and is inexpensive, although it does have temperature limitations. At low temperatures and above 400 ^{o}C, helium should be considered.

The range of operation is determined by the initial filling pressure.

Gas filled systems do provide a faster response than other filled devices, and as it converts temperature directly into pressure it is particularly useful in pneumatic systems. This type of measurement also has the advantage that there are no moving parts and no electrical stimulation.

The size of the bulb is not critical and in fact can be sized quite large for averaging measurements in large volumes such as dryers and ovens.

Mercury filled systems (Class V):

Mercury expansion systems are different from other liquid filled systems because of the properties of the metal. Mercury is toxic and can affect some industrial processes and for this reason is used less in filled systems. The high liquid density also limits on the elevation difference between the sensor and instrument.

Mercury filled systems provide the widest range of operation, which range from the freezing to boiling point of the metal, ie from -40 ^{o}C to 650 ^{o}C.

Systems using this technology are simple, inexpensive and have fast responses. Typically they also have good overrange protection.

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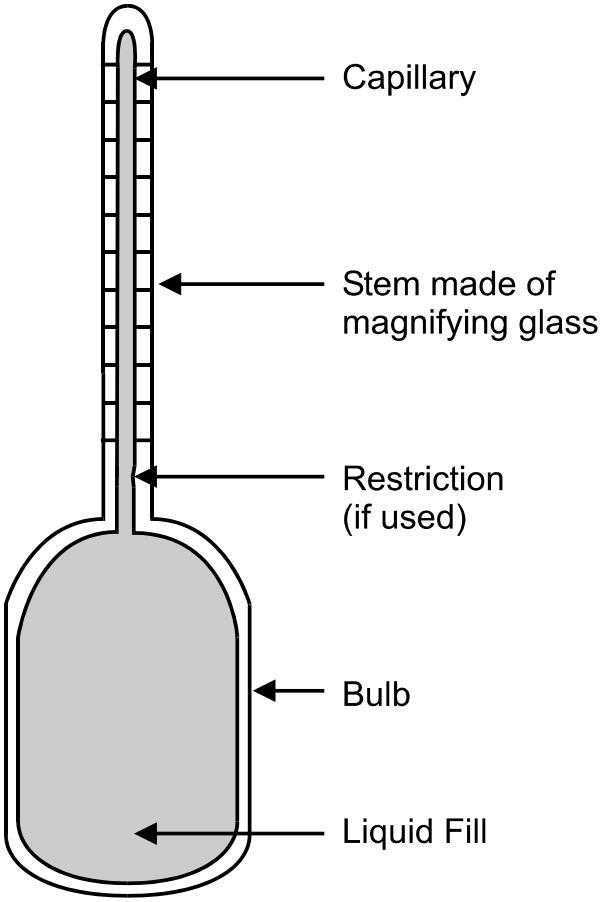


Figure 4.20
Liquid-in-glass thermometer

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Instruments for Filled Thermal System Sensors			
Type **Principle** **SAMA Class**	**Liquid** **Volume change** **I**	**Vapour (a)** **Pressure change** **II**	**Gas** **Pressure change** **III**
Fluids	Organic liquids (Hydro-carbons)	Organic liquids (Hydro-carbons), water	Pure gases
Lower range limit	-200°F (-130°C)	-425°F (-255°C)	-455°F (-270°C)
Upper range limit	+600°F (+315°C)	+600°F (+315°C)	+1,400°F (+760°C)
Narrowest span (b)	40°F (25°C)	70°F (40°C)	120°F (70°C)
Widest span	600°F (330°C)	400°F (215°C)	1,000°F (550°C)
Ambient temp.	IA full	Not required	
Compensation	IB case		IIIB case
Sensor size	Smallest	Medium	Largest
Typical sensor size for 100°C span	9.5mm (0.375 in) OD x 48mm (1.9in) long	9.5mm (0.375in.) OD x 50mm (2 in) long	22mm (7/8 in) OD x 70mm (6 in) long
Overrange capability	Medium	Least	Greatest
Sensor elevation effect	None	Class IIA, Yes Class IIB, None	None
Barometric pressure effect (altitude)	None	Slightly (greatest on small spans)	Slightly (greatest on small spans)
Scale uniformity	Uniform	Non-uniform	Uniform
Accuracy	±0.5 to ±1.0% of span	±0.5 to ±1.0% of span upper 2/3 of scale	±0.5 to ± 1.0% of span
Response (d) #1 Fastest #4 Slowest	#4	#1 – Class IIA #3 – Class IIB	#2
Cost	Highest	Lowest	Medium
Maximum standard	Class IA 30m or 100ft	30m or 100 ft	30m or 100 ft
Capillary length	Class IB 6, or 20ft		

Table 4.7
Instrument for filled thermal system sensors

Installation Techniques

To suit the differing applications, the temperature sensitive bulb is available in many different shapes and sizes. By using the largest bulb available for an application, this will limit temperature errors and allow smaller spans and longer capillaries.

A thermowell may need to be considered in adverse conditions and the associated problems need to be considered as the response time typically doubles when the thermowell is added. Response time is also affected by the size of the bulb, and doubles with the doubling of the bulb diameter.

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Stainless steel is commonly used for the bulb material, due to the range of temperatures it can operate within. Protection of the thin fragile capillary needs to be considered. Typically this may be protected using flexible armoured stainless steel or PVC covered bronze tubing. Use of an extension neck to the bulb means that the tubing is greater protected from the process material.

Typical Applications

Advantages

- Simple operation
- Robust
- Inexpensive
- No power source required
- Easily maintained
- Reasonably accurate

Disadvantages

- Bulky
- Slow response time
- Wide spans only
- Subject to gauge pressure problems
- Non linear

Application Limitations

The temperature that is displayed is a result of the pressure in the filled system. However the pressure in the system is a result of the temperature around the bulb and the temperature around the rest of the system.

Because the device is intended to measure process temperature only, temperature compensation may be required.

When the design is fully compensated for ambient temperature, the letter 'A' is added to the class distinction. If however only the case is compensated for, which may be sufficient, then the letter 'B' is added to the class distinction.

Vapour filled systems (Class II) do not require compensation as the liquid/vapour transition occurs within the bulb.

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Summary

This type of temperature measurement, as previously mentioned, relies on the measurement of pressure. As the pressure is referenced to the atmosphere, it is possible that atmospheric pressure changes can affect the pressure reading (also known as gauge pressure). This barometric change can cause an error of about 0.1% in the final measurement.

4.5.3 Bimetallic

Basis of Operation

Bimetallic temperature sensors work on the basis that different metals expand by different amounts. A bimetallic device consists of two metals bonded together which have different coefficients of expansion. Bending occurs as one metal expands more than the other. To amplify the mechanical movement of the deflection, the bimetallic device is generally wound into a spiral or helical form.

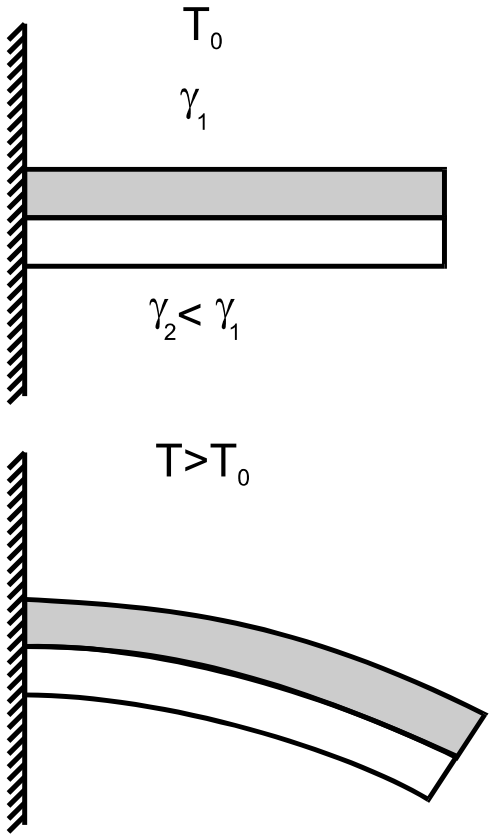


Figure 4.21
A bimetal strip will curve when exposed to a temperature change because of the differential thermal expansion coefficients

A bimetal strip will bend when exposed to a temperature variation. This is due to the different thermal coefficients of the two metals.

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Selection and Sizing

The operating ranges vary to cater for different applications, the lower range being from -70 ^{o}C to 50 ^{o}C, with a higher range available of up to 100 ^{o}C to 550 ^{o}C. Obviously, ranges are available between these extremes.

Installation Techniques

Bimetallic thermometers can be used in a thermowell. A thermowell has the added advantages of allowing the removal or replacement of the device without opening up the process tank or piping.

They are typically in the helical form when used in thermowells.

Vibration and heat transfer can be a problem with some applications. However, selecting a unit that is completely sealed can overcome these limitations. A dry gas is generally used in the dial face portion of the assembly while silicone fluid fills the stem and surrounds the coil. Having fluid around the coil can assist in mechanical damping and heat transfer.

Advantages

- Inexpensive
- Simple construction

Disadvantages

- Limited accuracy
- Indication or simple switching only
- Localised measurement only
- Easily decalibrated due to mechanical shock
- Hysteresis

Application Limitations

Primarily used for simple switching or indication on a dial. Use is generally restricted to local measurement only.

Summary

Bimetallics are used in numerous applications because of their inexpense and simple operation. Such applications range from household thermostats to HVAC equipment.

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4.6 Non Contact Pyrometers

Pyrometric methods of temperature measurement use the electromagnetic radiation that is emitted from a material. The emitted radiation is proportional to the temperature.

There are two basic types of pyrometers:
- Infrared pyrometers
- Acoustic pyrometers

4.6.1 Infrared Pyrometers

Basis of Operation

Any object with a temperature above absolute zero will radiate electromagnetic energy. Infrared pyrometers measure the amount of energy radiated from an object in order to determine its temperature.

Infrared or radiation pyrometers use an optical system to focus the radiated energy onto a sensing device.

Selection & Sizing

Different materials transmit radiation of different wavelengths. In selecting an effective radiation pyrometer, a wavelength-band needs to be chosen that is not transmitted by the material. Also the selected wavelength-band cannot be absorbed by the environment.

The pyrometer wavelength needs to be sized for the process.

There are a number of different types of infrared pyrometers:
- Total radiation
- Single wavelength
- Dual wavelength

Total radiation :

This type of radiation thermometer measures all the energy, over a broad spectrum, emitted by the object. Because of the broad spectrum of sensing, this type of measurement is easily affected by any impurities or inconsistencies between the sensor and the object. Greater errors occur for materials with lower emissivity.

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Single wavelength :

Unlike the broad spectrum, this type of radiation thermometer measures the magnitude of radiation at one wavelength. This optimises the accuracy of the device by selecting the wavelength best suited to the object being measured. This type of measurement provides good accuracy provided the emissivity is known.

Dual wavelength :

When the emissivity is low and hard to measure, or even variable, accuracy and reliability can be improved with a dual wavelength sensor. This type measures the magnitude of two wavelengths at the same time. The temperature is calculated from the two readings.

Typical Applications

Pyrometers are used for measuring high temperatures and where contact with the object is not possible.

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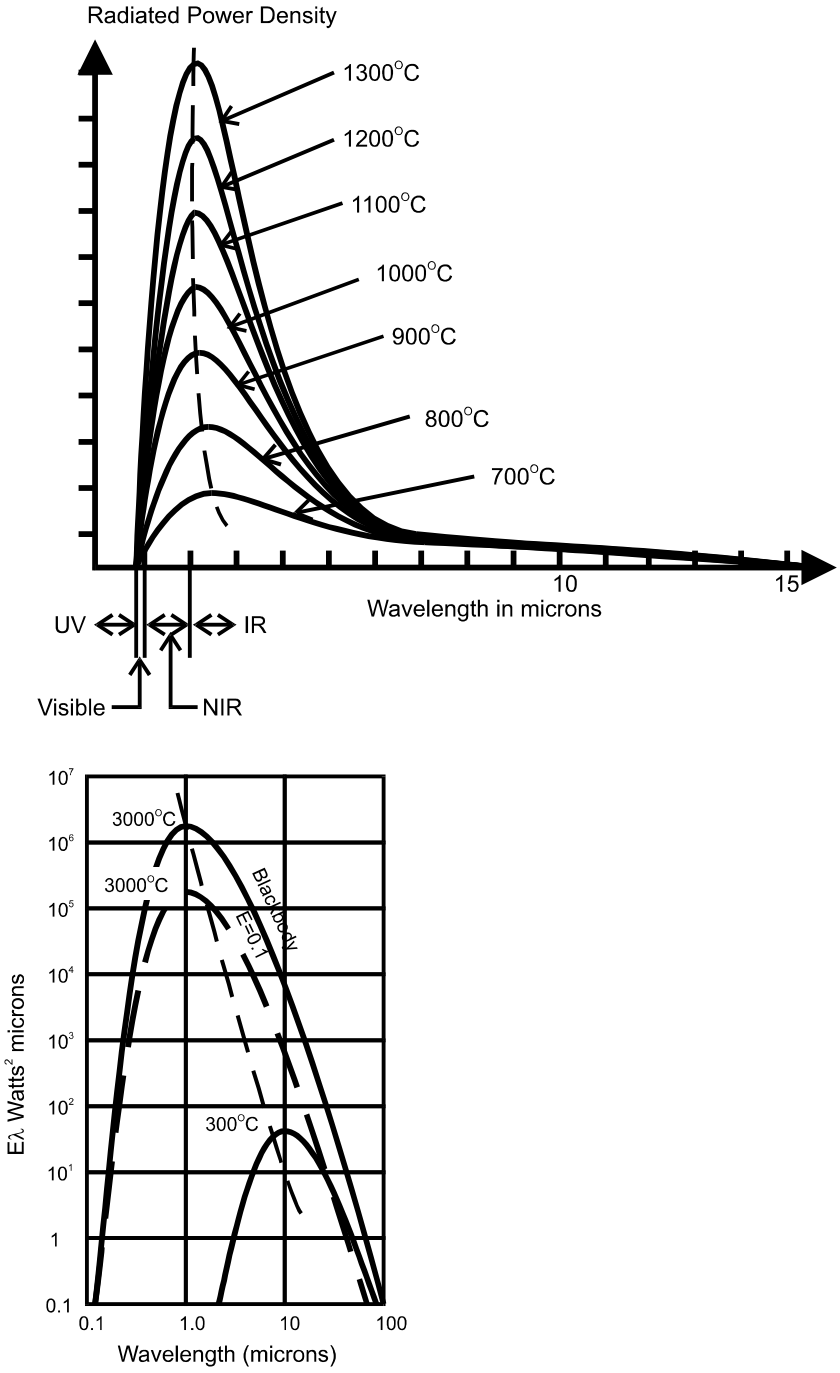


Figure 4.22
Any object above absolute zero will radiate electromagnetic energy. As the temperature increases, the peak moves to shorter wavelengths.

Advantages

- Non contact measurement
- High temperature sensing
- Remote sensing
- Fast response and can sense objects in motion
- Sense small or area targets

Disadvantages

- Expensive
- Non linear response
- Subject to emissivity of material
- Require wide range of operation

Application Limitations

Radiation pyrometry relies on thermal radiation leaving the material being measured. This is dependent on both the temperature of the object and the emittance of the material.

The accuracy is affected by reflections, presence of gases in the radiation path and the surface emissivity of the object. The surface emissivity can vary with wavelength, temperature and also the chemical composition of the object being measured. Other factors include the surface of the material, which include its texture or roughness and oxidation.

4.6.2 Acoustic Pyrometers

Basis of Operation

Acoustic pyrometers work on the principle that the speed of sound in a gas is dependant on the nature of the gas and its temperature. The time of flight is used, and since the distance between points is known it is possible to measure any change in conditions. This principle is adapted for liquid and solid temperature measurement also.

Typical Applications

Acoustic pyrometers are used when requiring an average temperature or the temperature over a large area or volume of gas.

Acoustic pyrometers are useful for measuring gas temperatures inside kilns and furnaces. They work over a very large temperature range and are useful for mapping thermal contours. Unfortunately cost is often prohibitive.

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4.7 Humidity

The moisture content in gases is termed humidity and is expressed as either:

- Relative humidity
- Absolute humidity
- Dew point

There are a vast number of methods and technologies developed over the years to measure humidity. Following is a list of such, with this section only focusing on the current and commonly used types.

Relative Humidity

- Mechanical
- Wet and dry bulb
- Surface resistivity devices
- Crystal frequency change

Absolute Humidity

- Gravimetry
- Electrolysis
- Infrared
- Conductivity
- Capacitance
- Colour change
- Karl Fischer titration
- RF power absorption
- Neutron reflection
- Heat of absorption or desorption
- Nuclear magnetic resonance

Dew Point Measurement

- Chilled mirror
- Lithium chloride
- Wet bulb thermometer

4.7.1 Capacitive

Capacitance is one of the most common forms of humidity measurement.

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Basis of Operation

Any type of capacitor requires two conductive plates separated by a di-electric. Humidity measurement is performed by exposing the dielectric to the moisture in the air. The moisture changes the relative capacitance between the two electrodes, which can be measured.

The metals used for the plates (or at least one plate) need to be porous in order for the dielectric to change in moisture content. The metals do vary as developments in technology progress.

The dielectric material is the single most important factor affecting the performance of capacitive humidity sensors. Other factors are the processing conditions, the sensor structure and the properties of the water permeable electrode or electrodes.

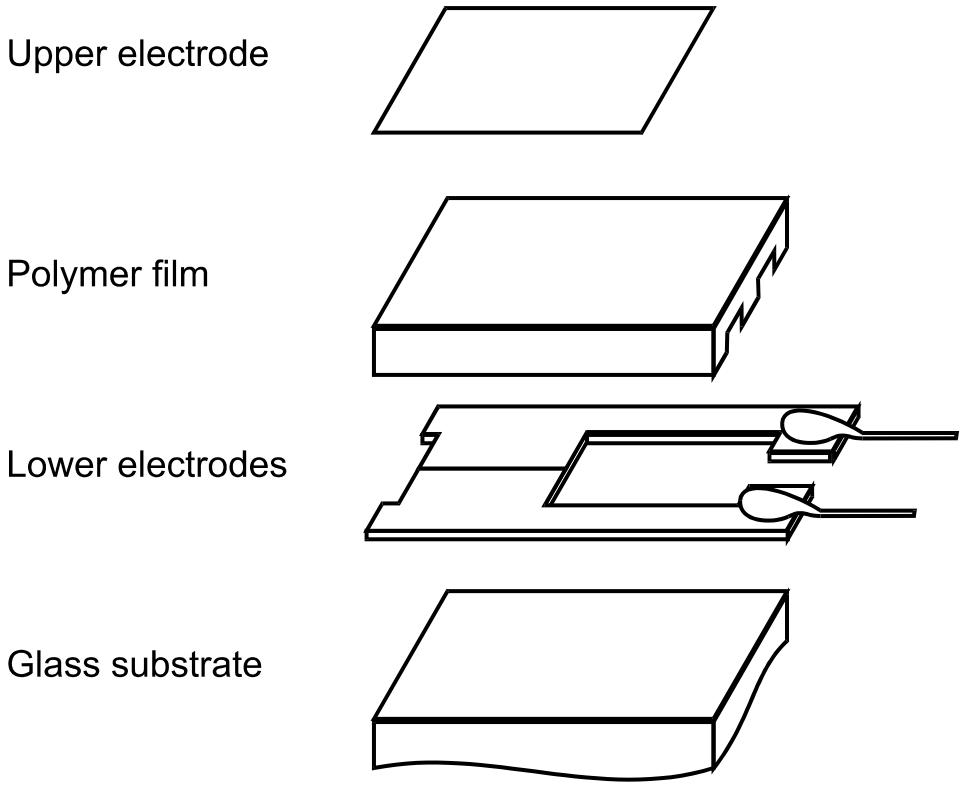


Figure 4.23
Structure of the Humicap® Sensor

Water permeable top electrodes:

The water permeable top electrode was originally thin gold film, but advances in technology have seen a cracked chromium film now used. Some of the advantages of the cracked chromium over the thin gold film are:

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- Easier to manufacture
- Good corrosion resistance
- Good adhesion to polymer films
- Faster response times

Porous metal films is another emerging technology that only allows water to permeate the film, excluding alcohols. This feature improves the stability of the sensor particularly in applications involving vapours, other than water vapour. This technology also enables an ultra-fast response.

Dielectric:

Aluminium oxide is commonly used as the dielectric medium. Doping of the aluminium oxide may be done with lithium chloride to give a greater operating range, typically down to -70 ^{o}C. The use of lithium chloride also requires that the device be individually calibrated, and slows the response.

Bottom electrode:

A conductive aluminium base material is typically used as the bottom electrode.

Typical Applications

Humidity sensors are typically used in building controls. The most effective location for a wall mounted sensor is near the return air duct, where the air leaving the room has the best averaged humidity.

Advantages

- Wide range of sensing
- Small size

Disadvantages

- Non linear
- Require periodic calibration
- Not highly accurate

Application Limitations

Capacitive humidity devices that use an aluminium oxide moisture sensor are not very accurate and each sensor requires a separate calibration curve, which is non-linear.

Periodic calibration is necessary because of errors due to contamination and general ageing of the device.

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Summary

This type of sensing is generally used for HVAC applications due to their low cost and relatively small size. In these applications, accuracy is not critical and periodic calibration can be performed without disturbing the smooth operation of the process.

4.8 Installation Considerations

4.8.1 Location and Cabling

The best installation practice is to mount all electronics directly on top of the thermowell. This eliminates lead-wire and noise effects. If, for some reason this cannot be done, the lead wires need to be twisted and shielded to maximise protection from electrical interference. The wires should also not be stressed, either under tension or from tight bends. Large diameter cabling should be considered to minimise electrical resistance and increase mechanical robustness.

4.8.2 Thermowells

A thermowell has the added advantages of allowing the removal or replacement of the sensing device without opening up the process tank or piping.

Thermowells need to be considered when installing temperature sensing equipment. The length of the thermowell needs to be sized for the temperature probe.

Consideration of the thermal response needs to be taken into account. If a fast response is required, and the sensor probe already has adequate protection, then a thermowell may impede system performance and response time. Note that when a thermowell is used the response time is typically doubled.

Thermowells can provide added protection to the sensing equipment, and can also assist in maintenance and period calibration of equipment.

Thermopaste assists in the fast and effective transfer of thermal dynamics from the process to the sensing element. Application and maintenance of this material needs to be considered. Regular maintenance and condensation can affect the operation of the paste.

4.8.3 Effects of Self Heating

Any electrical temperature sensing device can be prone to self heating. In measuring the resistance of a device, a current must be passed through it. When current passes through a resistance, energy is lost or dissipated in the form of heat. When the current is kept low, this heat loss is minimised and the effects are minimal.

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Other methods use the non-linear characteristics of the thermistors to rapidly switch the output for a temperature change. These thermistors can be used as thermostats for overheating protection.

4.8.4 Noise

One of the main problems, particularly with thermocouples, is their susceptibility to electrical noise. The voltages that are generated are in the order of a few millivolts. In an industrial environment it is not uncommon to have voltages of 100's of times greater of electrical noise generated by higher power equipment.

A thermocouple makes for an excellent antenna, as it picks up high frequency electromagnetic radiation.

The three most common noise reduction techniques are as follows:

1. Twist and shield (grounded foil sheath) the extension or lead wires from the thermocouple to the measurement system.

2. Ground the measurement junction at the point of measurement. The grounding is typically to the inside of the stainless steel sheath that covers the actual thermocouple.

3. Use a transmitter with very good common mode voltage rejection, and locate as close to the thermocouple as possible.

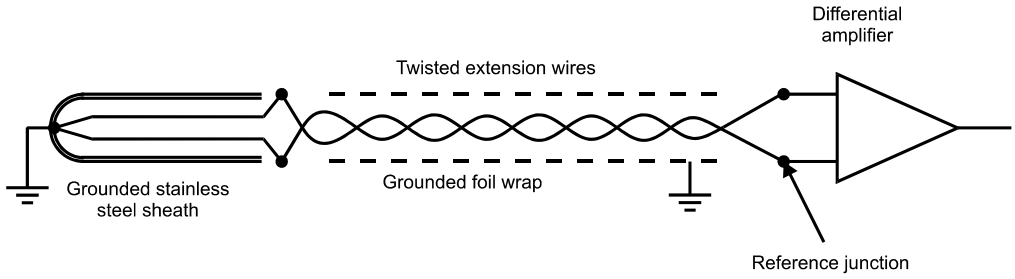


Figure 4.24
Protect the thermocouple signal against electrical noise by shielding, twisting and using a differential input.

The advantage of grounding the measurement junction is that the electrical noise is distributed equally on each wire of the thermocouple.

4.9 Impact on the Overall Control Loop

The time response ishow long it takes a system to come into thermal equilibrium with its surrounds. That is, how long it takes a sensor to read the actual temperature.

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The response of a thermocouple is related to the size of the wire and any protective material used with the sensor. Many industrial thermocouples use thick wire or are encased in stainless steel sheathing, and may have time constants as high as 10 to 20 seconds. Conversely, thermocouples made from very small gauge wire can have time constants as small as 10 to 20 milliseconds.

The response time can be specified under conditions of good and poor thermal contact, so the application environment can be accounted for.

4.10 Selection Tables

The figure below highlights the different responses between thermocouples, RTD's and thermistors.

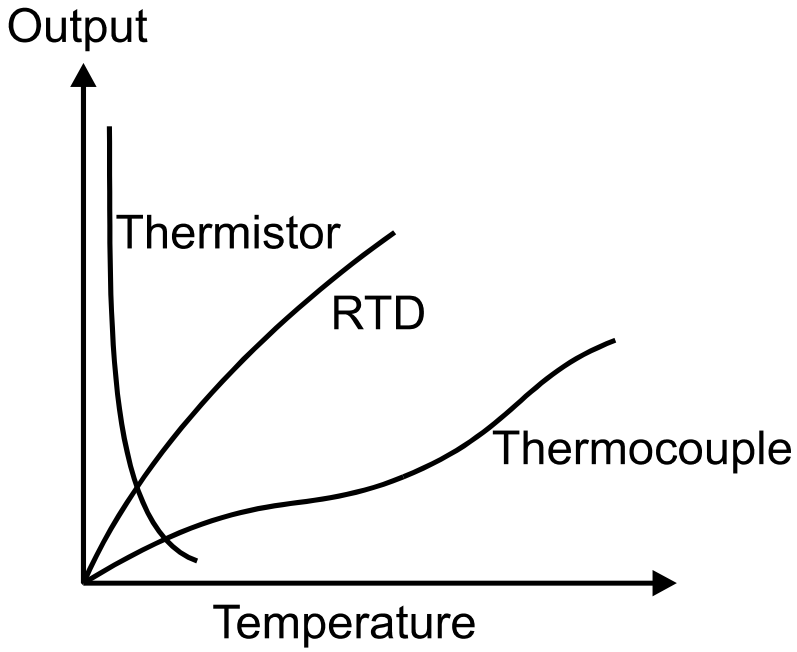


Figure 4.25
Thermistor versus RTD's versus Thermocouples

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Orientation table for temperature sensors								
Type	Temperature Range °F	Available Span	Accuracy % of span	Stability	Repeat-ability	Response Time	Sensit-ivity	Linear
T/Cs:								
Type T	-300 to 750	<100 to 1000	0.1	G	G	G	G	N
Type J	32 to 1500	<100 to >1000	0.1	G	G	G	G	N
Type K	-300 to 2300	<100 to >1000	0.1	G	G	G	G	-
Types R&S	32 to 3000	<1000 to >1000	0.1	G	G	E	E	N
RTDs:								
Pt	-400 to 1200	<100 to >1000	0.15	E	E	G	G,E	-
Nickel	-350 to 600	<100 to <1000	0.25	G,E	E	G	E	N
Thermistors	-100 to 500	<100 to <1000	0.2	F	G	E	E	N
Electronic	-350 to 325	<100 to <1000	0.2-2	G	G	G	E	-
Bimetallics	-100 to 1000	<100 to <1000	1-2	E	F	G	G	N
Filled:								
Liquid	-50 to 800	<100 to <1000	0.5-2	E	F	F	G	-
Vapour	0 to 600	<100 to <1000	0.5-2	E	F	G	F	N
Gas	-100 to 1800	<1000 to >1000	0.5-2	E	F	G	F	-
Mercury	-10 to 1000	<100 to >1000	0.5-2	E	F	F	G	-
Glass stem	-10 to 1000	<100 to <1000	0.1-2	E	E	F	G	-
Infrared	-10 to 6300	<1000 to >1000	1-2	F	F	E	G	N
Ultrasonic	1500 to 15000	<100 to >1000	5	G	F,G	E	E,G	N

Table 4.8
Orientation table for temperature sensors

4.11 Future Technologies

Electronic Temperature Sensors

Transistors or diodes can be used for temperature measurement. The output of such a device is very linear and is good for a temperature range of -55 to 150°C. These electronic devices are not as accurate as other devices, however they are inexpensive.

Sensor Size

Newer sensors use miniaturisation to maximise the number of applications and uses. This is not new. The miniaturisation of non-contact sensors is becoming more common with cost and specifications making them suitable in applications where contact sensors were previously used.

The development of smaller infrared sensors means that they have become more economical and maintain:

- calibrated accuracy
- minimal thermal drift
- stability

Applications of noncontact measurement is expanded with the use of fibre-optics, which does not require the need for miniaturisation. The fibre acts as a carrier for the

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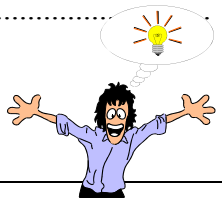
radiated energy. These have the added advantage of isolation of the components from direct heat or other hazardous surrounds.

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Tips and Tricks



Tips and Tricks



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Thermocouple Comparison

ISA Type designation	Positive wire	Negative wire	µV/°F	Recommended Temp limits °F	Scale linearity	Atmosphere recommended	Favourable points	Unfavourable points
B	Pt70-Rh30	Pt70-Rh6	3-6	32 to 3380	Same as type R	Inert or slow oxidising		
E	Chromel	Constantan	15-42	-300 to 1800	Good	Oxidising	Highest emf/°F	Larger drift than other base metal couples
J	Iron	Constantan	14-35	32 to 1500	Good; nearly linear from 300 to 800	Reducing	Most economical	Becomes brittle below 32°F
K	Chromel	Alumel	9-24	-300 to 2300	Good; most linear of all T/Cs	Oxidising	Most linear	More expensive than T or J
R	Pt87	Platinum	3-8	32 to 3000	Good at high temps	Oxidising	Small size	More expensive
S	Pt90-Th10	Platinum	3-7	32 to 3200	Same as R	Oxidising	Same as R	More expensive
T	Copper	Constantan	8-35	-300 to 750	Good but crowded at low end	Oxidising but reducing	Resists corrosion from moisture	Limited temperature range
Y	Iron	Constantan	22-33	-200 to 1800	About same as J	Reducing		Nonstandard
-	Tungsten	W74-Re26	1-12	0 to 4200	Same as R	Inert or vacuum	High temperature	Brittle; expensive
-	W94-Re6	W74-Re26	1-10	0 to 4200	Same as R	Inert or vacuum	Same as above	Slightly less brittle than above
-	Copper	Gold-cobalt	0.5-25	-450 to 0	Reasonable above 60°K		Good output at very low temp	Expensive lab-type T/C
-	Ir40-Rh60	Iridium	1-4	0 to 3800	Same as R	Inert		Brittle; expensive

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